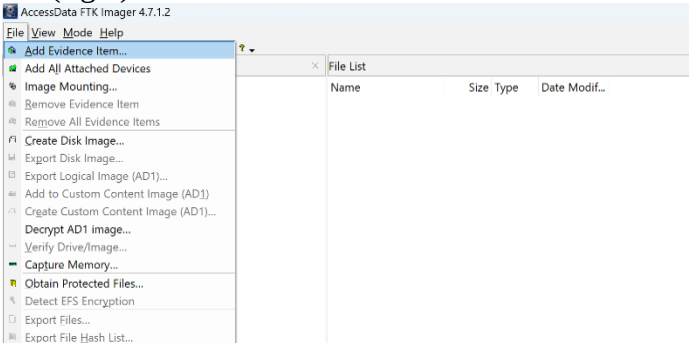
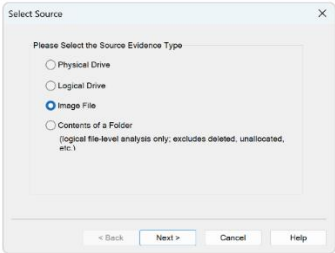
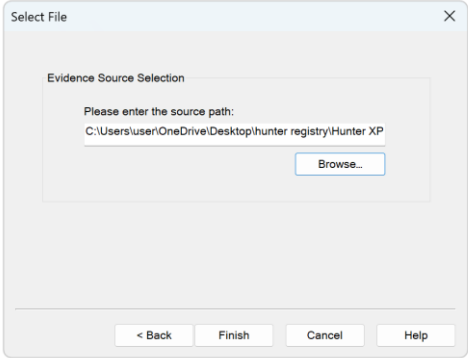
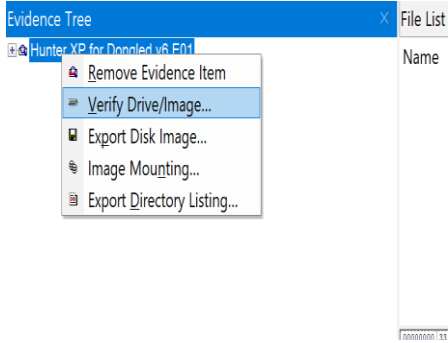


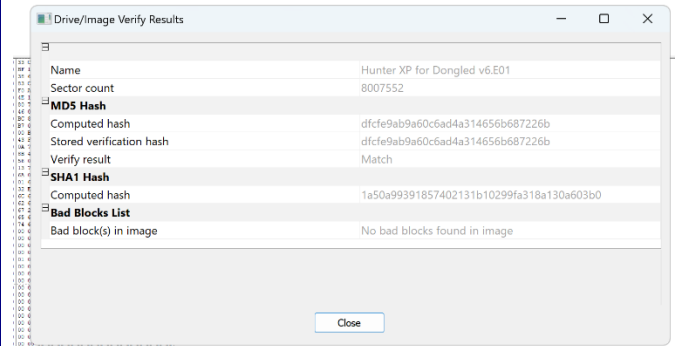
Contemporaneous Notes

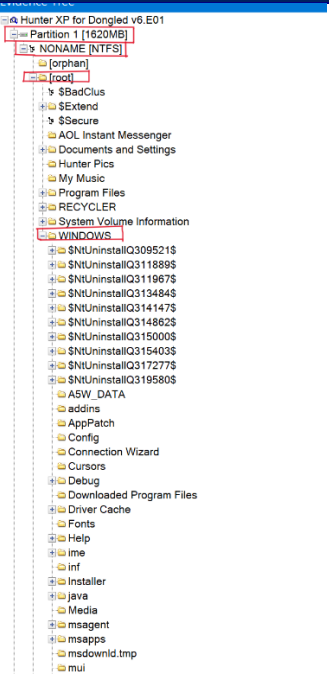
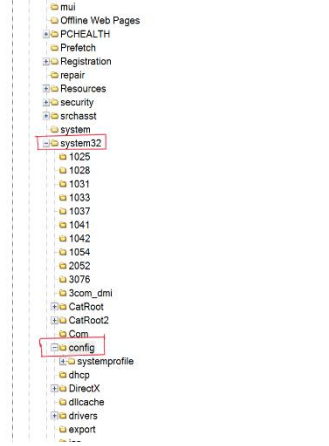
Note: If you decide to omit a process, then you should provide your reasons for doing so. You may add additional rows, as appropriate.

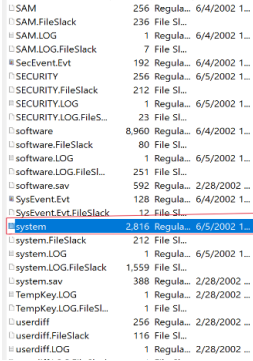
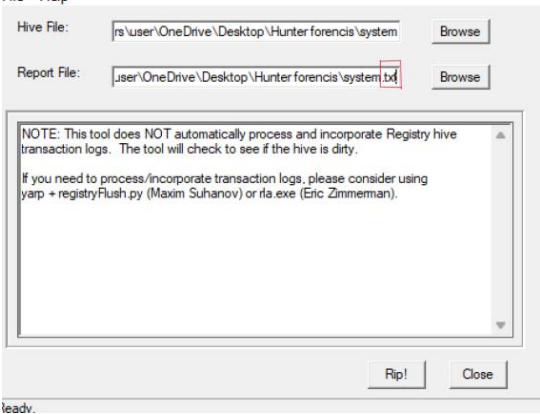
Examiner	Bibek Pandeya	Exam commenced	24 th November 2024 12:37 PM
Other relevant information	University ID: 24000863 The whole note is based on Nepali local time (NPT).  Path  Image/Screenshot	Software used, versions and licensing	All tools are open-sourced 1. Autopsy 4.19.2 2. AccessData FTK Imager 4.7.1.2 3. RegRipper 3.0 4. AccessData Registry Viewer 1.8.0.5 5. Kali Linux 2023.3 6. Ophcrack 3.8.0

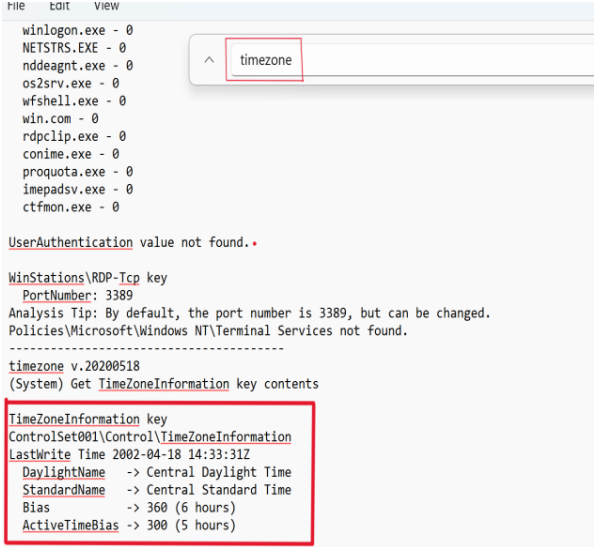
Action	Done?	Date	Time	Notes
Load case and verify image	Yes	24 th November 2024	1:26 PM	1. To load an image, first open the software AccessData FTK Imager and then add the image file you wish to investigate. Shown in the (fig.1):  Fig.1  Fig.2

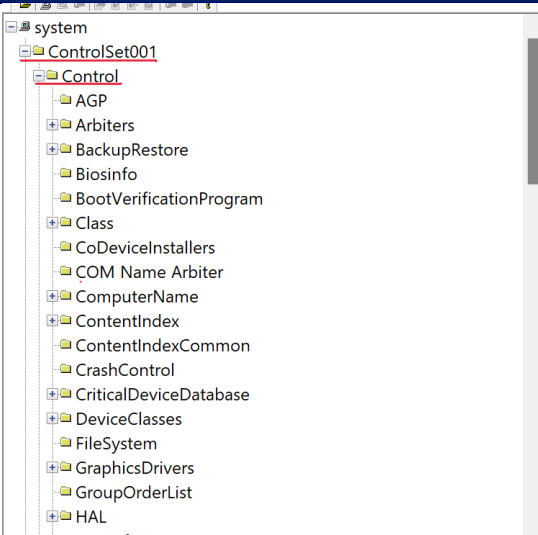
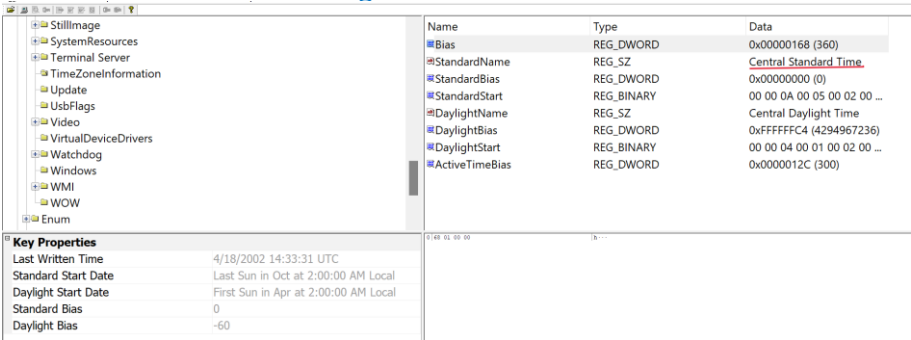
Action	Done?	Date	Time	Notes
				Since I already have an image for investigation, I need to select the image file from its appropriate location. (fig.2&3)  Fig.3 2. After successfully loading the image file, the next step is to verify its hash value. To do this, right-click on the image and select "Verify Drive/Image". (fig.4)  Fig.4

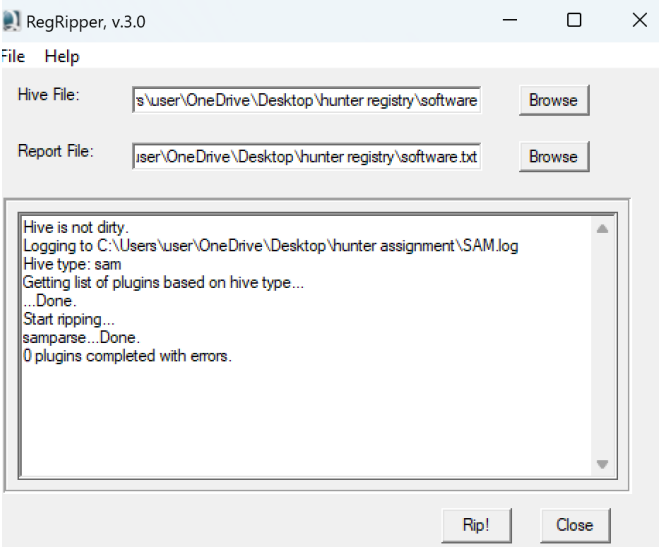
Action	Done?	Date	Time	Notes
				After clicking on the "Verify" option, the MD5 and SHA1 hash values of the image file will be displayed. (fig.5)  Fig.5
Load Case into second forensic tool for dual verification of at least 2 key artefacts, evidence items	Yes	24 th November 2024	2:45 PM	For this task we will use three tools AccessData Registry viewer 1.0.8.5, Reg ripper 3.0 & AccessData FTK Imager. To gather information about TimeZone & Operating system Details of suspect from the file SYSTEM & SOFTWARE. 1, TimeZone Using RegRipper 3.0 At first load the hunter xp image file to AccessData FTK Imager. Then, we search for the SAM, SECURITY, SOFTWARE, SYSTEM. LOCATION: Hunterxp/Partion 1/Noname/root/windows/system32/config

Action	Done?	Date	Time	Notes
				 Fig.6  Fig.7

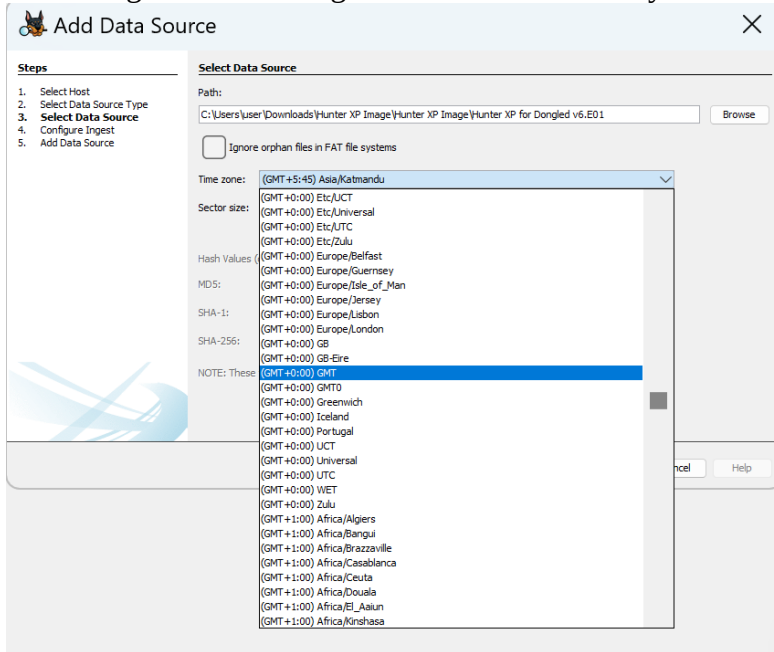
Action	Done?	Date	Time	Notes
				Now, after selecting Config search for SYSTEM because SYSTEM is the place where we can find Time Zone.  Fig.8 Now, after finding the system file export the file by right clicking to the file and giving the appropriate location. After exporting the file, input the file into the “REG RIPPER” software and create a report file in text format.  Fig.9 After clicking the rip option txt file appears in the same folder where TimeZone can be accessed by searching into the text file.

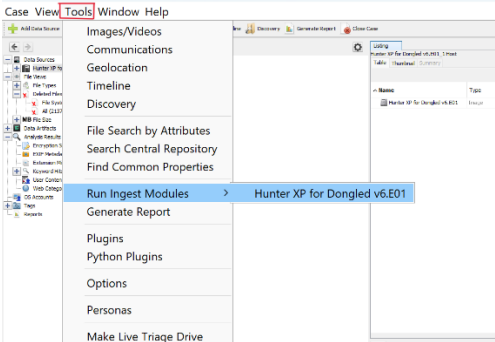
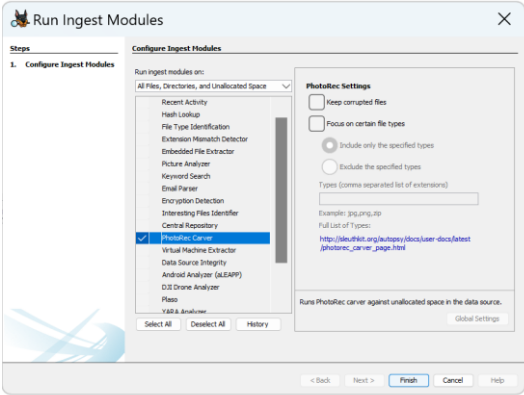
Action	Done?	Date	Time	Notes
				In the txt file we simply search for keyword TimeZone with the key “CTRL+F”.  The screenshot shows a text editor window with a list of running processes on the left and a search bar on the right. The search bar contains the text 'timezone'. Below the search bar, the results of the search are displayed, showing the 'TimeZoneInformation' key details. The details include the key name, the last write time, and the current time zone settings. Fig.10 1.1, TimeZone Using Registry Viewer 1.0.8.5 To know the TimeZone of The HunterXP file load the system file in registry viewer and navigate through location: System/controlset001/control/

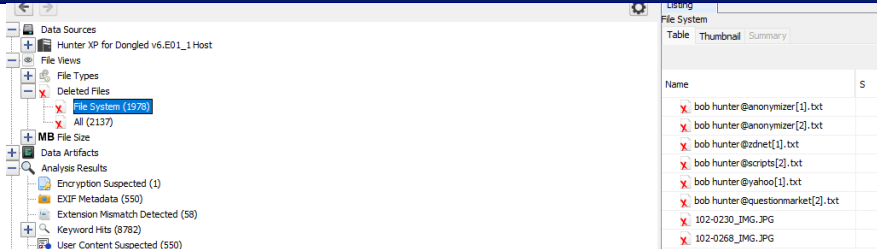
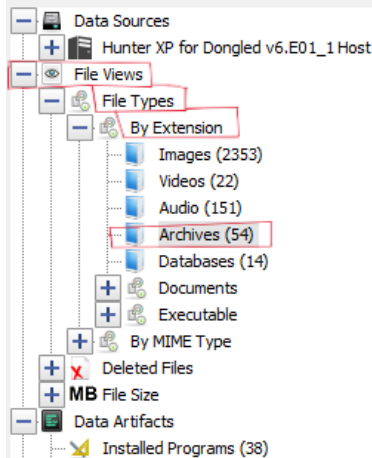
Action	Done?	Date	Time	Notes
				 Fig.11  Fig.12 2, operating System information Using Reg ripper 3.0 For details of the operating system, repeat the process of TimeZone analysis, but the difference is investigating the SOFTWARE file not the SAM or SYSTEM file.

Action	Done?	Date	Time	Notes
				Same as before after exporting the file from autopsy put the SOFTWARE file in the Reg ripper 3.0 tool to get the text file for the required information.  Fig.13 After searching the SOFTWARE file, it will show the details about operating system.

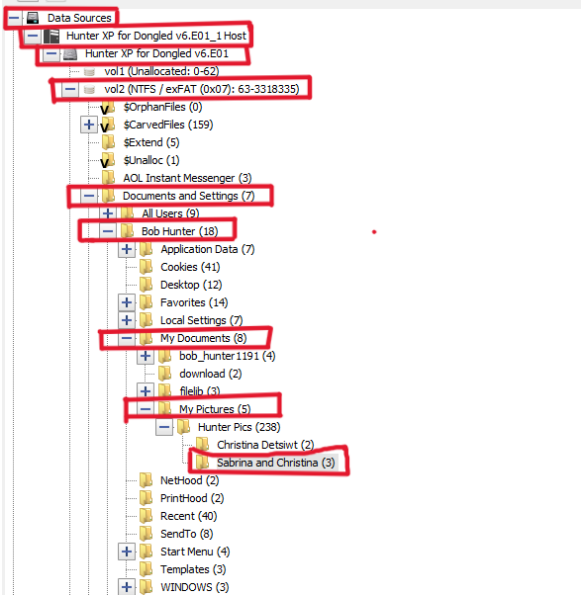
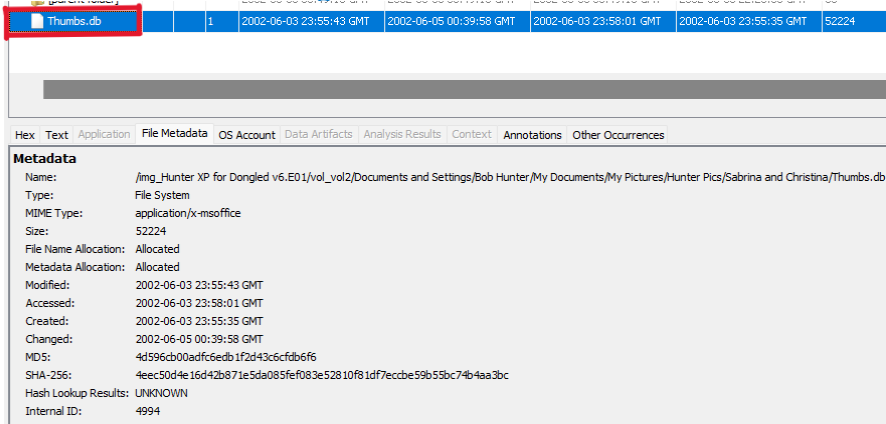
Action	Done?	Date	Time	Notes																																																						
				winver v.20200525 (Software) Get Windows version & build info ProductNameMicrosoft Windows XP BuildLab2600.xpclnt_qfe.010827-1803 RegisteredOrganizationPC User Company RegisteredOwnerPC User InstallDate2002-02-28 22:02:39Z wow64 v.20200515 (Software) Gets contents of WOW64\X86 key WOW64 Microsoft\WOW64\X86 not found. Microsoft\WOW64\arm not found. ----- wsh_settings v.20200517 (Software) Gets WSH Settings Microsoft\Windows Script Host\Settings Key LastWrite: 2002-02-28 21:52:14Z DisplayLogo1 ActiveDebugging1 SilentTerminate0 UseWTSAPFR1 Fig.14 2.1, Operating System Information Using Registry Viewer 1.0.8.5 To get the information first load the Software file exported From AccessData FTK Imager in “Registry Viewer 1.0.8.5”, then navigate through the location: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion CurrentVersion Accessibility AeDebug Asr Classes Compatibility Compatibility32 Console Drivers drivers.desc Drivers32 EFS Embedding Key Properties Last Written Time6/4/2002 23:01:57 UTC OS Install Date (UTC)Thu Feb 28 22:02:39 2002 OS Install Date (Local)Fri Mar 01 03:47:39 2002 <table><tr><th>Name</th><th>Type</th><th>Data</th></tr><tr><td>#CurrentBuild</td><td>REG_SZ</td><td>1.511.1.0 (Obsolete data ...)</td></tr><tr><td>#InstallDate</td><td>REG_DWORD</td><td>0x3C7EABFF (1014933759)</td></tr><tr><td>#ProductName</td><td>REG_SZ</td><td>Microsoft Windows XP</td></tr><tr><td>#RegDone</td><td>REG_SZ</td><td>(value not set)</td></tr><tr><td>#RegisteredOrganization</td><td>REG_SZ</td><td>PC User Company</td></tr><tr><td>#RegisteredOwner</td><td>REG_SZ</td><td>PC User</td></tr><tr><td>#SoftwareType</td><td>REG_SZ</td><td>SYSTEM</td></tr><tr><td>#CurrentVersion</td><td>REG_SZ</td><td>5.1</td></tr><tr><td>#CurrentBuildNumber</td><td>REG_SZ</td><td>2600</td></tr><tr><td>#BuildLab</td><td>REG_SZ</td><td>2600.xpclnt_qfe.010827-1...</td></tr><tr><td>#CurrentType</td><td>REG_SZ</td><td>Uniprocessor Free</td></tr><tr><td>#SystemRoot</td><td>REG_SZ</td><td>F:\WINDOWS</td></tr><tr><td>#SourcePath</td><td>REG_SZ</td><td>D:\386</td></tr><tr><td>#PathName</td><td>REG_SZ</td><td>F:\WINDOWS</td></tr><tr><td>#ProductId</td><td>REG_SZ</td><td>55277-005-6418583-216...</td></tr><tr><td>#DigitalProductId</td><td>REG_BINARY</td><td>A4 00 00 00 03 00 00 00 ...</td></tr><tr><td>#LicenseInfo</td><td>REG_BINARY</td><td>7E 49 71 40 6A 74 E1 95 ...</td></tr></table> Fig.15 Therefore, we can see both Artifacts exactly match with each other. We can see Fig (10 & 12) which show the same TimeZone details. Similarly, In the Fig (14 & 15) we can see similar result of operating system.	Name	Type	Data	#CurrentBuild	REG_SZ	1.511.1.0 (Obsolete data ...)	#InstallDate	REG_DWORD	0x3C7EABFF (1014933759)	#ProductName	REG_SZ	Microsoft Windows XP	#RegDone	REG_SZ	(value not set)	#RegisteredOrganization	REG_SZ	PC User Company	#RegisteredOwner	REG_SZ	PC User	#SoftwareType	REG_SZ	SYSTEM	#CurrentVersion	REG_SZ	5.1	#CurrentBuildNumber	REG_SZ	2600	#BuildLab	REG_SZ	2600.xpclnt_qfe.010827-1...	#CurrentType	REG_SZ	Uniprocessor Free	#SystemRoot	REG_SZ	F:\WINDOWS	#SourcePath	REG_SZ	D:\386	#PathName	REG_SZ	F:\WINDOWS	#ProductId	REG_SZ	55277-005-6418583-216...	#DigitalProductId	REG_BINARY	A4 00 00 00 03 00 00 00 ...	#LicenseInfo	REG_BINARY	7E 49 71 40 6A 74 E1 95 ...
Name	Type	Data																																																								
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#RegisteredOrganization	REG_SZ	PC User Company																																																								
#RegisteredOwner	REG_SZ	PC User																																																								
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#SourcePath	REG_SZ	D:\386																																																								
#PathName	REG_SZ	F:\WINDOWS																																																								
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#DigitalProductId	REG_BINARY	A4 00 00 00 03 00 00 00 ...																																																								
#LicenseInfo	REG_BINARY	7E 49 71 40 6A 74 E1 95 ...																																																								

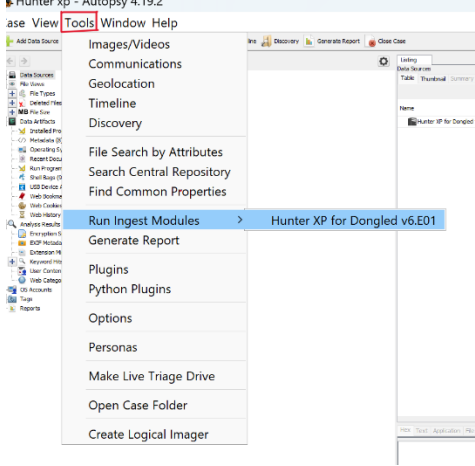
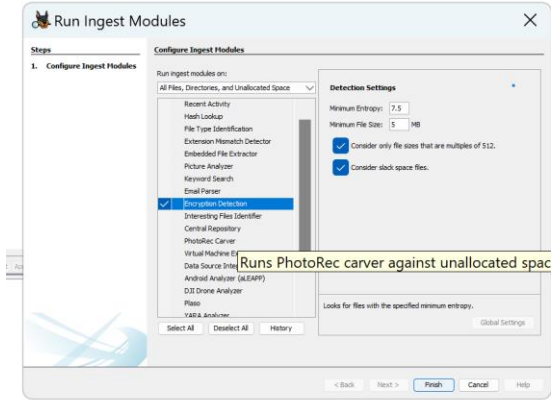
Action	Done?	Date	Time	Notes
Time Zone Adjusted? Report Time Zone used for Analysis.	Yes	24 th November 2024	3:35 PM	Yes, we adjusted Time zone of the Hunter Xp image file as per the UK local time which is (GMT +0:00) GMT Because we are examining the file as assignment and the university is at UK. (fig.16)  Fig.16 The Actual Time zone is (GMT +5:45) Asia/Kathmandu but we adjusted the time zone to (GMT +0:00) GMT Because as I mentioned above, we are student at UK university.

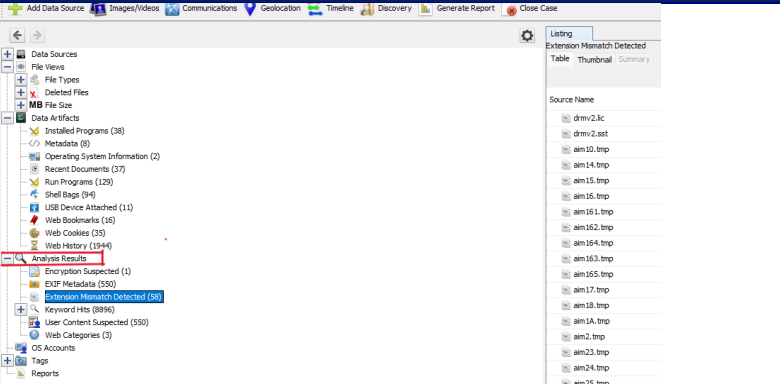
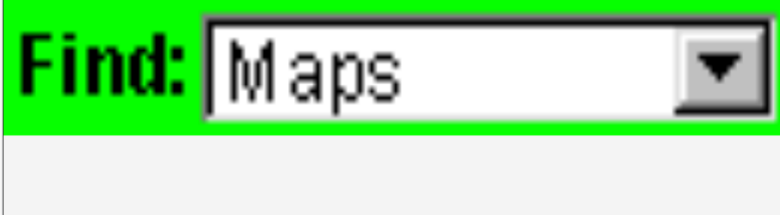
Action	Done?	Date	Time	Notes
Recover lost folders (NTFS, FAT16&32).	Yes	24 th November 2024	4:03 Pm	Just after following the directions below, we can easily recover deleted folders. (fig.17) 1. Select the Ingest Module for "PhotoRec Carver" to recover all deleted files. 2. Navigate to the Deleted Files or Unallocated Space folder. 3. PATH: Tool/RunningIngestModule/HunterXP/PhotoRec Carver.  Fig.17  Fig.18 From here, we can see all deleted files a total of 2137 (fig.19).

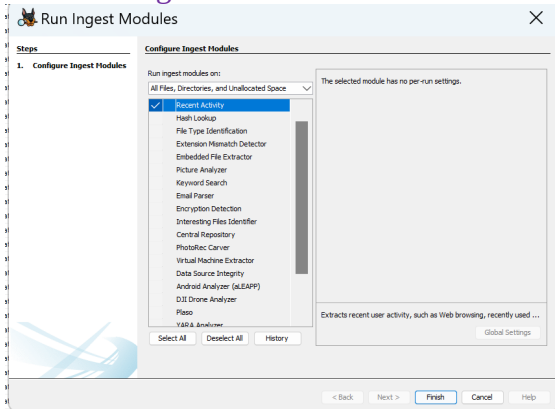
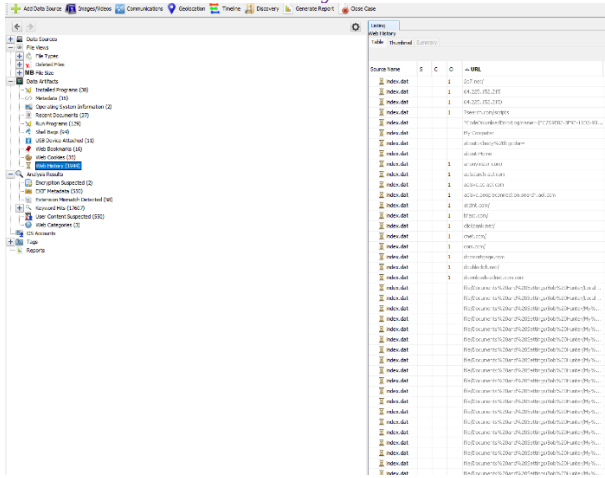
Action	Done?	Date	Time	Notes
				 Fig.19
Mount archives; zip, thumbs.db, etc.	Yes	24 th November 2024	5:12 Pm	To see the Zip Files, Navigate through the location(fig.20) Fileviews/Filetypes/Byextension/Archives  Fig.20 Here, the file whose name ends with .zip are the Zip files, (fig.21)

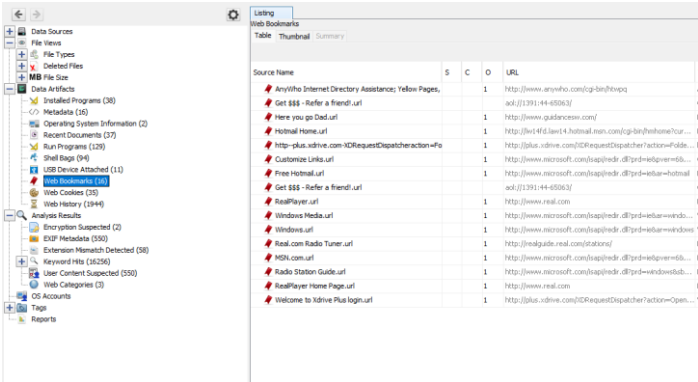
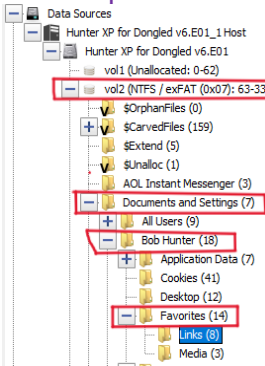
Action	Done?	Date	Time	Notes																																																																																													
				ase Listing Archives Table Thumbnail Summary <table><tr><th>Name</th><th>S</th><th>C</th><th>O</th><th>Modified Time</th><th>Change Time</th></tr><tr><td>jaws.jar</td><td></td><td></td><td>0</td><td>2000-08-04 22:06:02 GMT</td><td>2002-03-01 1</td></tr><tr><td>rt.jar</td><td></td><td>▼</td><td>0</td><td>2000-08-04 22:06:02 GMT</td><td>2002-03-01 1</td></tr><tr><td>sunrsasign.jar</td><td></td><td></td><td>0</td><td>2000-08-04 22:05:50 GMT</td><td>2002-03-01 1</td></tr><tr><td>raclass.zip</td><td>✗</td><td></td><td></td><td>0000-00-00 00:00:00</td><td>0000-00-00 0</td></tr><tr><td>ident[1].cab</td><td>✗</td><td></td><td></td><td>0000-00-00 00:00:00</td><td>0000-00-00 0</td></tr><tr><td>Df389.zip</td><td>✗</td><td>▼</td><td></td><td>2002-06-03 21:15:33 GMT</td><td>2002-06-03 2</td></tr><tr><td>Df789.zip</td><td>✗</td><td>▼</td><td></td><td>2002-06-03 21:15:33 GMT</td><td>2002-06-03 2</td></tr><tr><td>Df935.zip</td><td>✗</td><td>▼</td><td></td><td>2002-06-03 21:15:33 GMT</td><td>2002-06-03 2</td></tr><tr><td>lucert[1].cab</td><td></td><td></td><td>1</td><td>2002-06-05 00:05:34 GMT</td><td>2002-06-05 0</td></tr><tr><td>yinst[1].cab</td><td></td><td></td><td>0</td><td>2002-05-14 16:11:17 GMT</td><td>2002-05-14 1</td></tr><tr><td>luident[1].cab</td><td></td><td></td><td>1</td><td>2002-06-05 00:05:33 GMT</td><td>2002-06-05 0</td></tr><tr><td>ident.cab</td><td></td><td></td><td>1</td><td>2002-05-28 10:30:49 GMT</td><td>2002-06-04 2</td></tr><tr><td>ident.cab</td><td></td><td></td><td>1</td><td>2002-05-28 10:30:49 GMT</td><td>2002-06-03 1</td></tr><tr><td>wuaucomp.cab</td><td></td><td></td><td>1</td><td>2002-01-21 23:27:12 GMT</td><td>2002-06-03 1</td></tr></table> yinst[1].cab Fig.21 ✗ f0004964.html.gz	Name	S	C	O	Modified Time	Change Time	jaws.jar			0	2000-08-04 22:06:02 GMT	2002-03-01 1	rt.jar		▼	0	2000-08-04 22:06:02 GMT	2002-03-01 1	sunrsasign.jar			0	2000-08-04 22:05:50 GMT	2002-03-01 1	raclass.zip	✗			0000-00-00 00:00:00	0000-00-00 0	ident[1].cab	✗			0000-00-00 00:00:00	0000-00-00 0	Df389.zip	✗	▼		2002-06-03 21:15:33 GMT	2002-06-03 2	Df789.zip	✗	▼		2002-06-03 21:15:33 GMT	2002-06-03 2	Df935.zip	✗	▼		2002-06-03 21:15:33 GMT	2002-06-03 2	lucert[1].cab			1	2002-06-05 00:05:34 GMT	2002-06-05 0	yinst[1].cab			0	2002-05-14 16:11:17 GMT	2002-05-14 1	luident[1].cab			1	2002-06-05 00:05:33 GMT	2002-06-05 0	ident.cab			1	2002-05-28 10:30:49 GMT	2002-06-04 2	ident.cab			1	2002-05-28 10:30:49 GMT	2002-06-03 1	wuaucomp.cab			1	2002-01-21 23:27:12 GMT	2002-06-03 1			0
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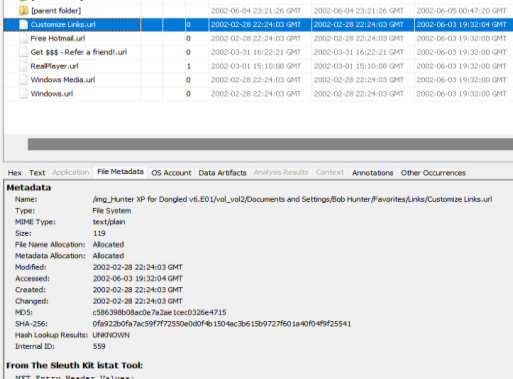
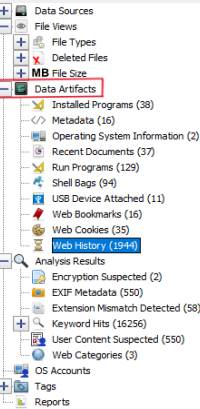
Action	Done?	Date	Time	Notes
				 The screenshot shows a file system tree with several folders and files. The path highlighted in red is: Data Sources > Hunter XP for Dongled v6.E01_1 Host > Hunter XP for Dongled v6.E01 > vol1 (Unallocated: 0-62) > vol2 (NTFS / exFAT (0x07): 63-3318335) > \$OrphanFiles (0) > \$CarvedFiles (159) > \$Extend (5) > \$Unalloc (1) > AOL Instant Messenger (3) > Documents and Settings (7) > All Users (9) > Bob Hunter (18) > Application Data (7) > Cookies (41) > Desktop (12) > Favorites (14) > Local Settings (7) > My Documents (8) > bob_hunter1191 (4) > download (2) > filelib (3) > My Pictures (5) > Hunter Pics (238) > Christina Detlwt (2) > Sabrina and Christina (3) > Thumbs.db.  The screenshot shows the metadata for a file named 'Thumbs.db'. The file is located at the path: /img_Hunter XP for Dongled v6.E01/vol2/Documents and Settings/Bob Hunter/My Documents/My Pictures/Hunter Pics/Sabrina and Christina/Thumbs.db. The file is a File System application/x-msoffice file, 52224 bytes in size. It was modified on 2002-06-03 23:55:43 GMT, accessed on 2002-06-03 23:58:01 GMT, created on 2002-06-03 23:55:35 GMT, and changed on 2002-06-05 00:39:58 GMT. The MD5 hash is 4d596cb00adfc8edb1f2d43c5cfd6b6f6, the SHA-256 hash is 4eec50d4e16d42b871e5da085fe083e52810f81df7ecbce59b55bc74b4aa3bc, and the internal ID is 4994.

Action	Done?	Date	Time	Notes
File signature analysis (any interesting file mismatch?); Compute hash values (enable entropy computation)	Yes	25 th November 2024	11:48 Am	To see any file mismatch select run ingest module after that execute the encryption detection option as shown below:(fig.25)  Fig.25  Fig.26 After going through ingest module navigate through Analysis result and extension mismatch option.

Action	Done?	Date	Time	Notes
				 Fig.27  Fig.28

Action	Done?	Date	Time	Notes
Internet History, favourites, etc. Other browsers?	Yes	25 th November 2024	1:08 Pm	(fig.31) First, run the ingest module for information about internet history: Tools/runingestmodule/recent activities/  Fig.31 After that we navigate through: Data artifacts/web history  Fig.32

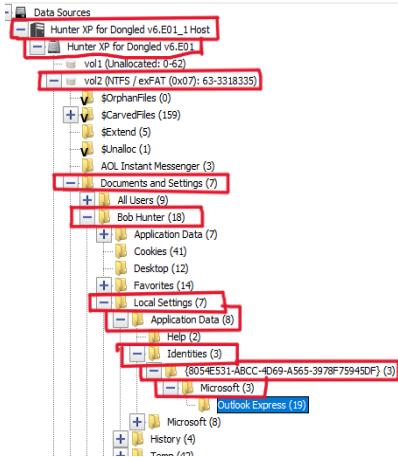
Action	Done?	Date	Time	Notes
				These are the websites or internet history which are accessed by the suspect “bob hunter”. In this Index.dat file we can see a lot of websites visited by the suspect. 2, Bookmarks, Favourites These are the links or URL the suspect has bookmarked(fig.33). Bookmarks:  Fig.33 For the favourites need to go to this location:(fig.34) Hunterxp/vol2/Documents and setting/Bob hunter/favourites  Fig.34

Action	Done?	Date	Time	Notes
				 The screenshot shows a table of file access events with columns for file name, size, and timestamps. Below the table, the 'Metadata' tab is selected, displaying details for the file 'Customize Links.url', including its type, size, and various timestamps.
				The browser used by the suspect is Internet Explorer: To see all the websites accessed using the browser navigate through the location: DataArtifacts/webhistory/  The screenshot shows a tree view of data artifacts. The 'Web History' folder is highlighted, indicating the location where web browsing history is stored.

Action	Done?	Date	Time	Notes																																																																																																																																																																																																																																							
				<table><tr><th>Source Name</th><th>S</th><th>C</th><th>O</th><th>URL</th><th>Date Accessed</th><th>Referrer URL</th><th>Program Name</th><th>Domain</th><th>Username</th><th>Data Source</th></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>207.net/</td><td>2002-06-03 15:26:12 GMT</td><td></td><td>Internet Explorer</td><td>207.net</td><td>Cookie-bob hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>64.225.152.215/</td><td>2002-06-03 13:07:25 GMT</td><td></td><td>Internet Explorer</td><td>64.225.152.215</td><td>Bob-Hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>64.225.152.215/</td><td>2002-06-03 13:07:22 GMT</td><td></td><td>Internet Explorer</td><td>64.225.152.215</td><td>Cookie-bob hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>7search.com/scripts</td><td>2002-06-04 18:49:18 GMT</td><td></td><td>Internet Explorer</td><td>7search.com</td><td>Cookie-bob hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td></td><td></td><td>TCODEDownloadErrorLogName=PC704E02-3F9E-11D3-93...</td><td>2002-05-14 13:01:05 GMT</td><td></td><td>Internet Explorer</td><td></td><td></td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td></td><td></td><td>My Computer</td><td>2002-06-03 11:40:46 GMT</td><td></td><td>Internet Explorer</td><td></td><td>Bob-Hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td></td><td></td><td>about:csdy%20google</td><td>2002-06-03 10:57:10 GMT</td><td></td><td>Internet Explorer</td><td></td><td>Bob-Hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td></td><td></td><td>about:home</td><td>2002-02-09 16:37:55 GMT</td><td></td><td>Internet Explorer</td><td></td><td>Bob-Hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>anonymizer.com/</td><td>2002-06-04 18:56:27 GMT</td><td></td><td>Internet Explorer</td><td>anonymizer.com</td><td>Cookie-bob hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>adsearh.aol.com</td><td>2002-06-04 18:30:27 GMT</td><td></td><td>Internet Explorer</td><td>aol.com</td><td>Bob-Hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>aolinc.cc.aol.com</td><td>2002-06-04 18:37:47 GMT</td><td></td><td>Internet Explorer</td><td>aol.com</td><td>Bob-Hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>aolinc.peopleconnection.search.aol.com</td><td>2002-06-04 18:29:54 GMT</td><td></td><td>Internet Explorer</td><td>aol.com</td><td>Bob-Hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>atdnt.com/</td><td>2002-06-03 12:45:17 GMT</td><td></td><td>Internet Explorer</td><td>atdnt.com</td><td>Cookie-bob hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>bfast.com/</td><td>2002-06-04 18:49:18 GMT</td><td></td><td>Internet Explorer</td><td>bfast.com</td><td>Cookie-bob hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>clickbank.net/</td><td>2002-06-04 18:49:17 GMT</td><td></td><td>Internet Explorer</td><td>clickbank.net</td><td>Cookie-bob hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>cnet.com/</td><td>2002-06-04 18:34:30 GMT</td><td></td><td>Internet Explorer</td><td>cnet.com</td><td>Cookie-bob hunter</td><td>Hunter SP for Dorgled v6.E01</td></tr><tr><td>index.dat</td><td></td><td>1</td><td></td><td>com.com/</td><td>2002-06-04 18:31:31 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Fig.37

Fig.37

Action	Done?	Date	Time	Notes
Emails, local and web based.	Yes	25 th November 2024	2:01 Pm	1, First, the web-based email used by suspect was performed by Hotmail platform. To know where the location of that file is containing that information to go to this location: /Img_Hunter XP for Dongled v6.E01/vol_vol2/Documents and Settings/Bob Hunter/Local Settings/Application Data/Identities/{8054E531-ABCC-4D69-A565-3978F75945DF}/Microsoft/Outlook Express/Bank Information.dbx  Fig.38 We can see in the below image that suspects are using “Hotmail.msn.com” for sending and receiving emails.

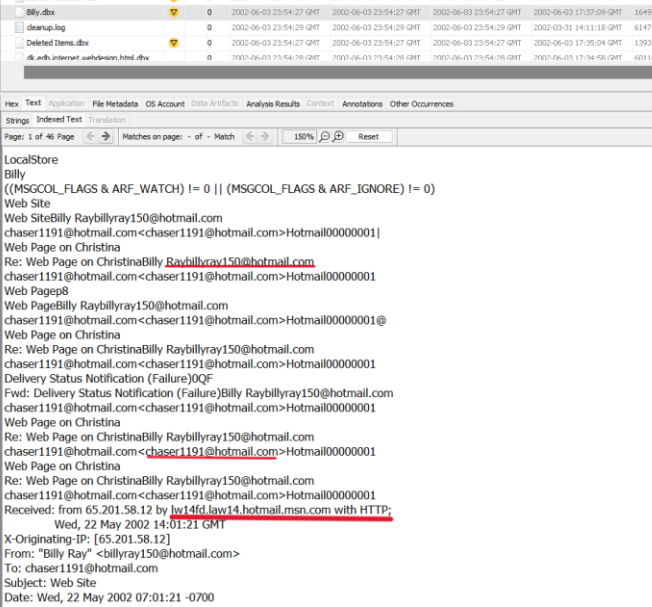
Action	Done?	Date	Time	Notes
				 Fig.39 2, Again, the suspects also used local email named “outlook express 6”. To go to the location where the file containing information is saved, just go through this location: /Img_Hunter XP for Dongled v6.E01/vol_vol2/Documents and Settings/Bob Hunter/Local Settings/Application Data/Identities/{8054E531-ABCC-4D69-A565-3978F75945DF}/Microsoft/Outlook Express/Deleted Items.dbx After arriving at this location, the suspects used “Microsoft outlook express team”. Furthermore, below is the image where suspect trying to setup or create the outlook express 6 email.

Fig.39

/Img_Hunter XP for Dongled v6.E01/vol_vol2/Documents and Settings/Bob Hunter/Local Settings/Application Data/Identities/{8054E531-ABCC-4D69-A565-3978F75945DF}/Microsoft/Outlook Express/Deleted Items.dbx

Action	Done?	Date	Time	Notes
				Deleted Items.dbx02002-06-03 23:54:27 GMT2002-06-03 23:54:27 GMT2002-06-03 23:54:27 GMT2002-06-03 17:35:04 GMT dk.edb.internet.webdesign.html.dbx02002-06-03 23:54:28 GMT2002-06-03 23:54:28 GMT2002-06-03 23:54:28 GMT2002-06-03 17:34:58 GMT HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences StringsIndexed TextTranslation Page: 1 of 1 PageMatches on page: - of - Match150%Reset LocalStorage Deleted Items ((MSGCOL_FLAGS & ARF_WATCH) != 0 (MSGCOL_FLAGS & ARF_IGNORE) != 0) <u>Welcome to Outlook Express 6</u> Welcome to Outlook Express 6Microsoft Outlook Express Teammsoe@microsoft.com New Outlook Express User From: "Microsoft Outlook Express Team" <msoe@microsoft.com> To: "New Outlook Express User" Subject: Welcome to Outlook Express 6 Date: <u>Sun, 31 Mar 2002 08:10:47 -0600</u> MIME-Version: 1.0 Content-Type: text/html; charset="iso-8859-1" Content-Transfer-Encoding: quoted-printable X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000 <HTML> Fig.40 Using this local mail one suspect had sent an image file to another, which can be clearly seen in below image. Inbox.dbx02002-06-03 23:54:27 GMT2002-06-03 23:54:27 GMT2002-06-03 23:54:27 GMT2002-03-31 14:10:47 GMT60116Alt Offline.dbx02002-06-03 23:54:29 GMT2002-06-03 23:54:29 GMT2002-06-03 23:54:29 GMT2002-03-31 14:10:46 GMT9656Alt Outbox.dbx02002-06-03 23:54:27 GMT2002-06-03 23:54:27 GMT2002-06-03 23:54:27 GMT2002-03-31 14:16:44 GMT1038196Alt HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences StringsIndexed TextTranslation Page: 1 of 60 PageMatches on page: - of - Match150%Reset LocalStorage Outbox ((MSGCOL_FLAGS & ARF_WATCH) != 0 (MSGCOL_FLAGS & ARF_IGNORE) != 0) Emailing: 103-0356_IMG.zip Emailing: 103-0356_IMG.zipWe Can Get Her!pay@us.comp billy@aol<bray@aol.com>Hotmail000000001 From: "We Can Get Her!" <pay@us.com> To: "billy@aol" <bray@aol.com> Subject: Emailing: <u>103-0356_IMG.zip</u> Date: Mon, 3 Jun 2002 16:17:28 -0500 MIME-Version: 1.0 Content-Type: multipart/mixed; boundary="-----_NextPart_000_0005_01C20B1A.27981A10" X-Priority: 3 X-MSMail-Priority: Normal X-Mailer: <u>Microsoft Outlook Express 6.00.2600.0000</u> X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000 This is a multi-part message in MIME format. -----_NextPart_000_0005_01C20B1A.27981A10 Content-Type: multipart/alternative; boundary="-----_NextPart_001_0006_01C20B1A.27981A10" -----_NextPart_001_0006_01C20B1A.27981A10 Content-Type: text/plain Fig.41

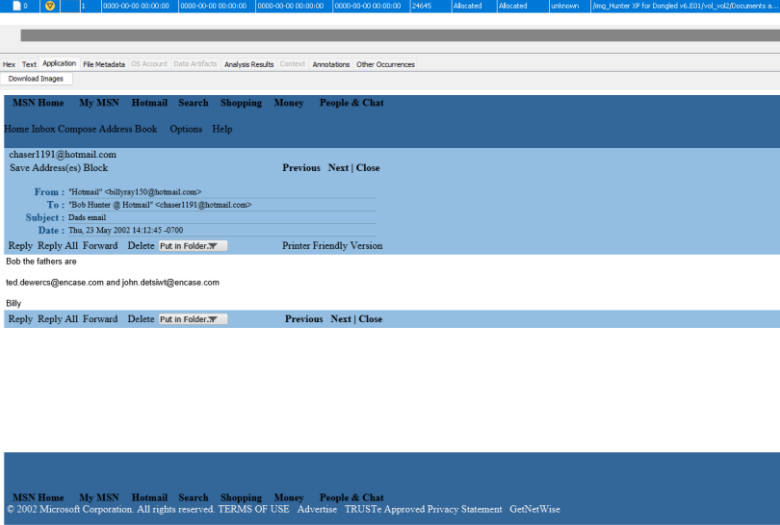
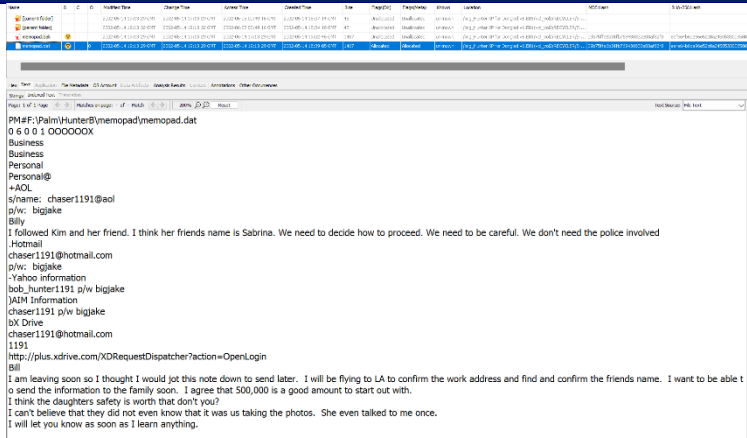
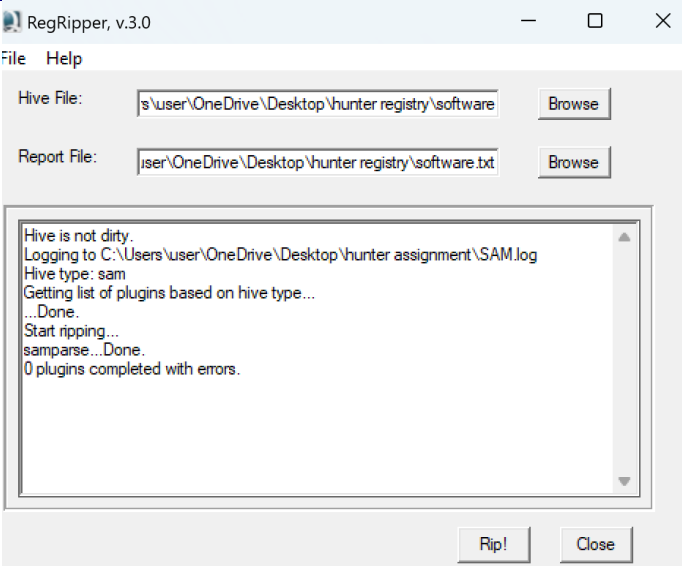
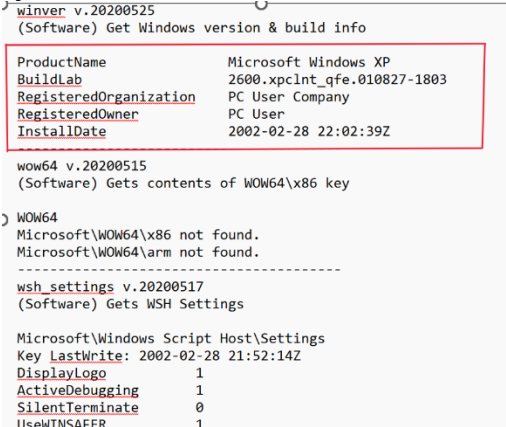
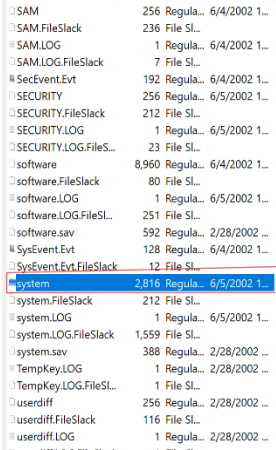
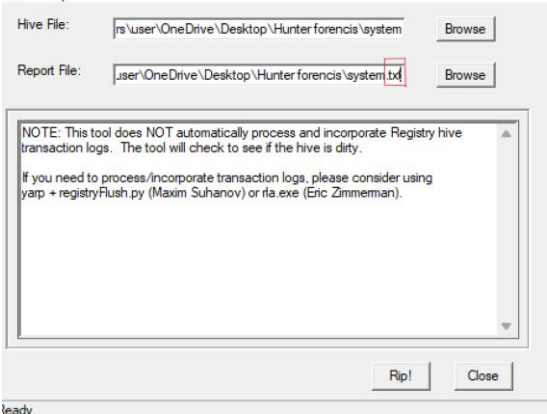
Action	Done?	Date	Time	Notes
				# Suspect's Operating Email to transfer information Below is the image where one suspect is providing the email of victims father email to another suspect.  The screenshot shows an email client interface. At the top, there's a navigation bar with links like 'MSN Home', 'My MSN', 'Hotmail', 'Search', 'Shopping', 'Money', and 'People & Chat'. Below this is a sub-header 'Home Inbox Compose Address Book Options Help'. The main content area displays an email from 'chaser1191@hotmail.com' to 'chaser1191@hotmail.com' with the subject 'Dads email'. The email body contains the text 'Bob the fathers are'. The interface includes various action buttons like 'Reply', 'Reply All', 'Forward', 'Delete', and 'Put in Folder'.

Fig.42

After this we can see the suspects talking about the amount of money to ask to the father.

Action	Done?	Date	Time	Notes
				 Fig.43
Retrieve operating system information, accounts information, software, time zone information etc.).	Yes	25 th November 2024	3:47 Pm	First, load the hunter xp image file to AccessData FTK Imager as we had done before. Then, search for the SAM, SECURITY, SOFTWARE, SYSTEM, LOCATION: Hunterxp/Partition 1/Noname/root/windows/system32/config 1, operating System information For details of the operating system, repeat the process of TimeZone analysis, but the difference is examine the SOFTWARE file not the SAM or SYSTEM file. Same as before after exporting the file from autopsy we put the SOFTWARE file in the Reg ripper 3.0 tool to get the text file for the required information.

Action	Done?	Date	Time	Notes
				 RegRipper, v.3.0 File Help Hive File: s:\user\OneDrive\Desktop\hunter registry\software Browse Report File: user\OneDrive\Desktop\hunter registry\software.txt Browse Hive is not dirty. Logging to C:\Users\user\OneDrive\Desktop\hunter assignment\SAM.log Hive type: sam Getting list of plugins based on hive type... ...Done. Start ripping... samparse...Done. 0 plugins completed with errors. Rip! Close Fig.44 After searching the SOFTWARE file, there are details of the operating system. <pre>winner v.20200525 (Software) Get Windows version & build info ProductName Microsoft Windows XP BuildLab 2600.xpclnt_qfe.010827-1803 RegisteredOrganization PC User Company RegisteredOwner PC User InstallDate 2002-02-28 22:02:39Z wow64 v.20200515 (Software) Gets contents of WOW64\x86 key WOW64 Microsoft\WOW64\x86 not found. Microsoft\WOW64\arm not found. ----- wsh_settings v.20200517 (Software) Gets WSH Settings Microsoft\Windows Script Host\Settings Key LastWrite: 2002-02-28 21:52:14Z DisplayLogo 1 ActiveDebugging 1 SilentTerminate 0 HkswTNSAFRR 1</pre> Fig.45

Action	Done?	Date	Time	Notes
				After that for TimeZone Information Choose SYSTEM file and execute the same process.  Fig.46  Fig.47 After clicking the rip option, there is a txt file where we need to search for Time Zone. In the txt file simply search for keyword TimeZone with the key “CTRL+F”.

Action	Done?	Date	Time	Notes
				File Edit View winlogon.exe - 0 NETSTRS.EXE - 0 nddeagnt.exe - 0 os2srv.exe - 0 wfshell.exe - 0 win.com - 0 rdpclip.exe - 0 conime.exe - 0 proquota.exe - 0 imepadsv.exe - 0 ctfmon.exe - 0 ^ timezone UserAuthentication value not found. WinStations\RDP-Tcp key PortNumber: 3389 Analysis Tip: By default, the port number is 3389, but can be changed. Policies\Microsoft\Windows NT\Terminal Services not found. ----- timezone v.20200518 (System) Get TimeZoneInformation key contents TimeZoneInformation key ControlSet001\Control\TimeZoneInformation LastWrite Time 2002-04-18 14:33:31Z DaylightName -> Central Daylight Time StandardName -> Central Standard Time Bias -> 360 (6 hours) ActiveTimeBias -> 300 (5 hours)

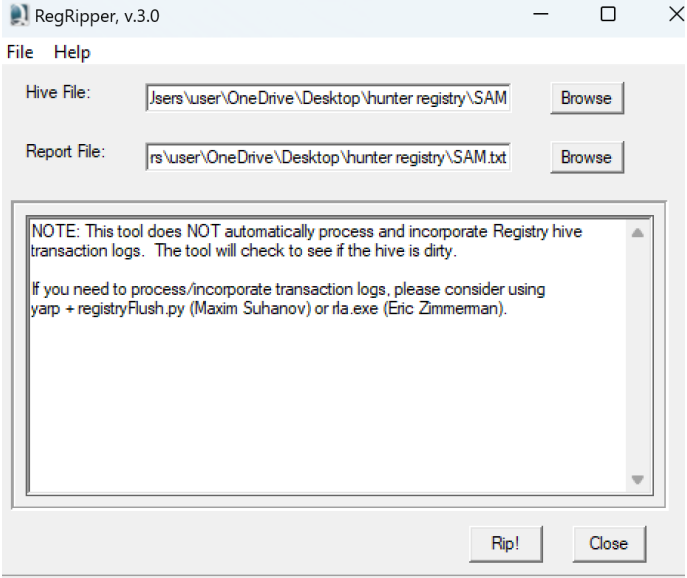
Fig.48

As a result, “Central Daylight Time” found which is the local time of North America.

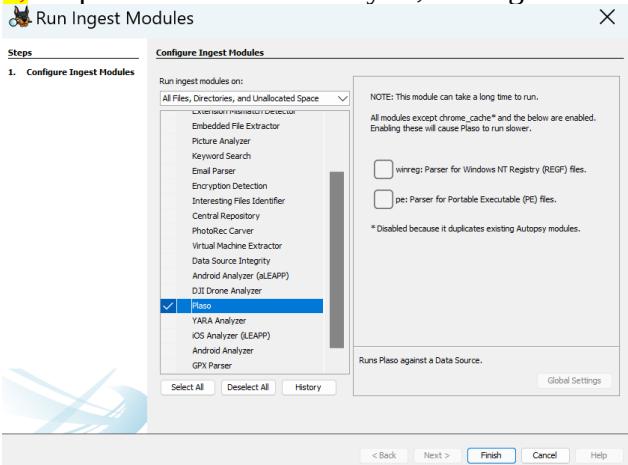
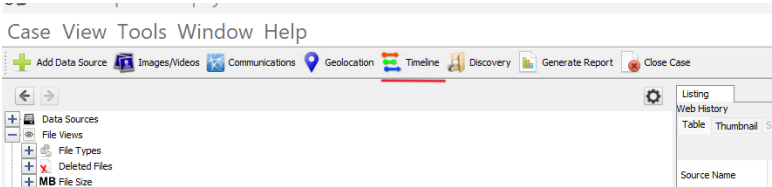
The suspect executed the crime from North America.

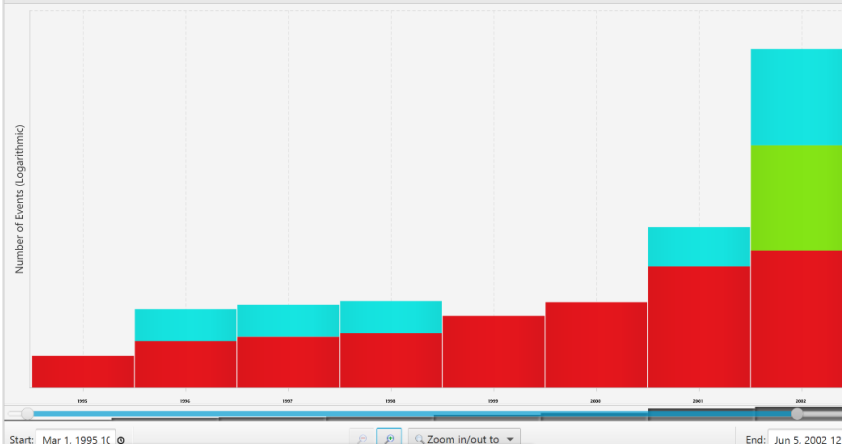
2, Account details/User details

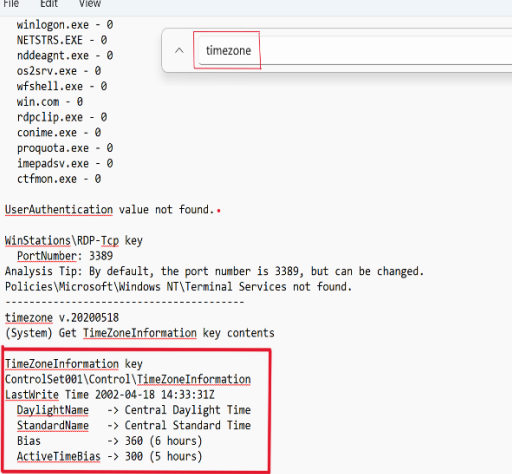
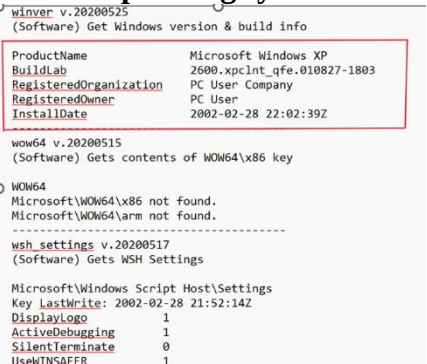
For user details, copy all the process used for TimeZone analysis, but the difference is for this task investigate the SAM not the SYSTEM.

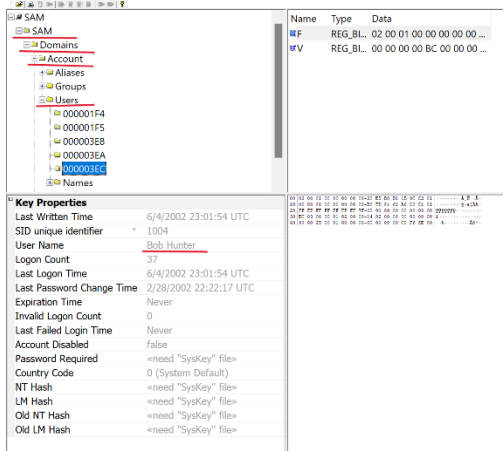
Action	Done?	Date	Time	Notes
				 Fig.49 Same as before, browse the SAM file and insert it in the Forensics Tool after then text file of SAM is generated where we get the information about the accounts. After searching for the user details, some default information and the suspects information (BOB HUNTER) were clearly seen. Below are the images of Default and the suspect's account information. The first three are default user and the Bob hunter is the suspect.

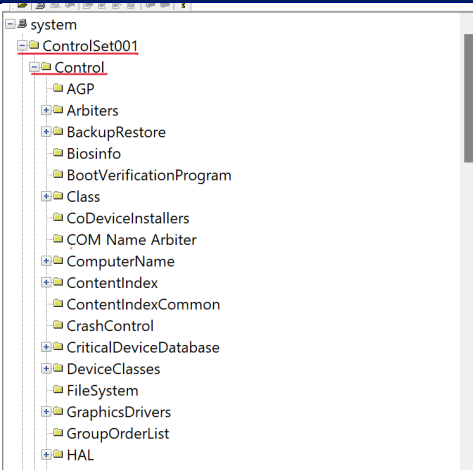
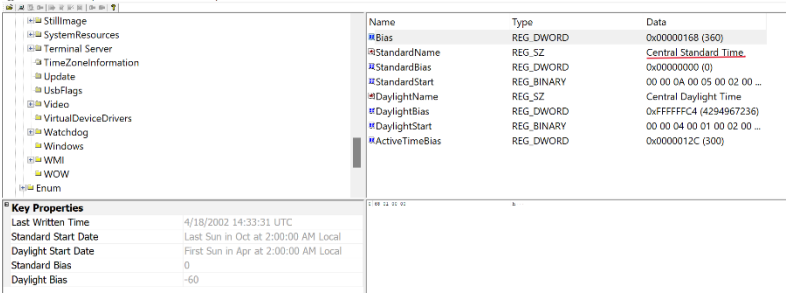
Action	Done?	Date	Time	Notes
				Hive (C:\Users\user\OneDrive\Desktop\hunter_assignment\SAM) is not dirty. samparse v.20220921 (SAM) Parse SAM file for user & group mbrshp info <pre> User Information ----- Username : Administrator [500] SID : S-1-5-21-1229272821-1580818891-854245398-500 Full Name : User Comment : Built-in account for administering the computer/domain Account Type : Default Admin User Account Created : Thu Feb 28 15:22:36 2002 Z Name : Last Login Date : Never Pwd Reset Date : Never Pwd Fail Date : Never Login Count : 0 --> Password does not expire --> Normal user account </pre> <pre> Username : Guest [501] SID : S-1-5-21-1229272821-1580818891-854245398-501 Full Name : User Comment : Built-in account for guest access to the computer/domain Account Type : Default Guest Acct Account Created : Thu Feb 28 15:22:36 2002 Z Name : Last Login Date : Mon Jun 3 16:49:37 2002 Z Pwd Reset Date : Never Pwd Fail Date : Never Login Count : 0 --> Password does not expire --> Password not required --> Normal user account </pre> Fig.50 <pre> Username : SUPPORT_388945a0 [1002] SID : S-1-5-21-1229272821-1580818891-854245398-1002 Full Name : CN=Microsoft Corporation,L=Redmond,S=Washington,C=US User Comment : This is a vendor's account for the Help and Support Service Account Type : Custom Limited Acct Account Created : Thu Feb 28 21:56:13 2002 Z Name : Last Login Date : Never Pwd Reset Date : Thu Feb 28 21:56:13 2002 Z Pwd Fail Date : Never Login Count : 0 --> Password does not expire --> Account Disabled --> Normal user account </pre> <pre> Username : Bob Hunter [1004] SID : S-1-5-21-1229272821-1580818891-854245398-1004 Full Name : User Comment : Account Type : Default Admin User Account Created : Thu Feb 28 22:22:17 2002 Z Name : Last Login Date : Tue Jun 4 23:01:54 2002 Z Pwd Reset Date : Thu Feb 28 22:22:17 2002 Z Pwd Fail Date : Never Login Count : 37 --> Password does not expire --> Password not required --> Normal user account </pre> Fig.51

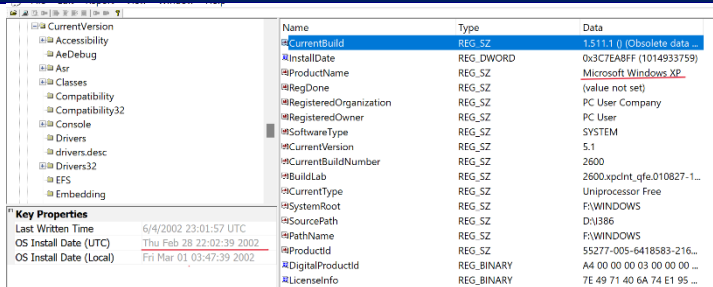
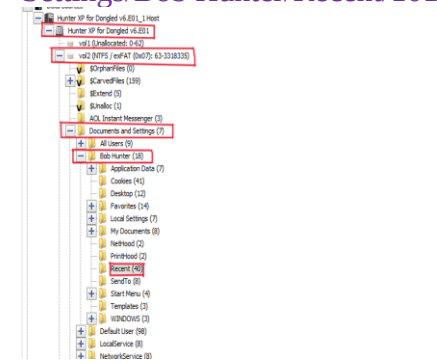
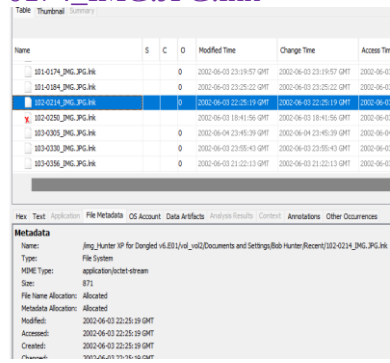
Action	Done?	Date	Time	Notes
Timeline analysis- Note date of last activity on the computer. System profiling.	Yes	26 th November 2024	1:19 Pm	1. To perform Timeline analysis, run ingest module and select plaso.  Fig.52 After that timeline section can be seen at the top containing the date and time information.  Fig.53 Finally, after analysing the timeline, we can say that the date of last activity on the computer was Jun 5, 2002, 12:50:12 AM.

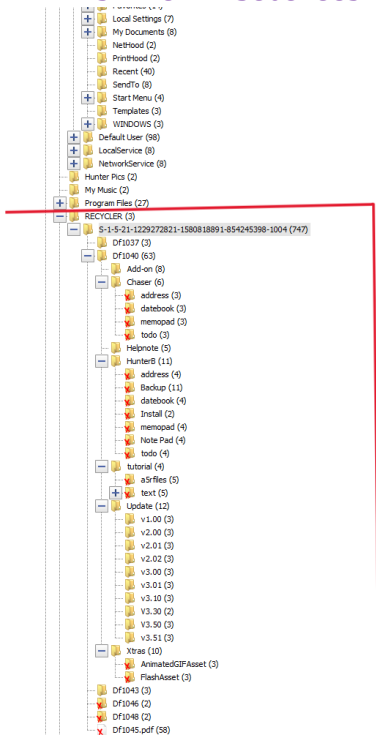
Action	Done?	Date	Time	Notes
				 Start: Mar 1, 1995 10 End: Jun 5, 2002 12 Fig.54 2, Now, for System profiling user details, TimeZone and operating system details play vital role. As, already done this before I will just put the outputs: 1. User Details We can see below the information about the suspect “bob hunter”. <pre>Username : SUPPORT_388945a0 [1002] SID : S-1-5-21-1229272821-1580818891-854245398-1002 Full Name : CN=Microsoft Corporation\Redmond,S=Washington,C=US User Comment : This is a vendor's account for the Help and Support Service Account Type : Custom Limited Acct Account Created : Thu Feb 28 21:56:13 2002 Z Name : Last Login Date : Never Pwd Reset Date : Thu Feb 28 21:56:13 2002 Z Pwd Fail Date : Never Login Count : 0 --> Password does not expire --> Account Disabled --> Normal user account</pre> <pre>Username : Bob Hunter [1004] SID : S-1-5-21-1229272821-1580818891-854245398-1004 Full Name : User Comment : Account Type : Default Admin User Account Created : Thu Feb 28 22:22:17 2002 Z Name : Last Login Date : Tue Jun 4 23:01:54 2002 Z Pwd Reset Date : Thu Feb 28 22:22:17 2002 Z Pwd Fail Date : Never Login Count : 37 --> Password does not expire --> Password not required --> Normal user account</pre> Fig.55

Action	Done?	Date	Time	Notes
				2. TimeZone  3. Operating system information 

Action	Done?	Date	Time	Notes
Registry analysis and Registry protected area	Yes	26 th November 2024	2:10 Pm	To analyse the registry of HunterXP image use registry viewer to see User Details, TimeZone Analysis & Operating System Details. 1. User Details (SAM FILE) First, export the file from autopsy of SAM, SYSTEM, SOFTWARE, SECURITY. Then, load the SAM file in the registry viewer and navigate through the location: Sam/domain/accounts/user  Fig.58 2. TimeZone (SYSTEM FILE): To know the TimeZone of The HunterXP file load the system file in registry viewer and navigate through location: System/controlset001/control/

Action	Done?	Date	Time	Notes
				 Fig.59  Fig.60 3. Operating System Information Using Registry Viewer 1.0.8.5 To get the information first load the Software file exported, then navigate through the location: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion

Action	Done?	Date	Time	Notes
				 Fig.61
Link files and Recycle Bin	Yes	26 th November 2024	3:08 Pm	1, link files To view the details of link files to go to this location: /Img_Hunter XP for Dongled v6.E01/vol_vol2/Documents and Settings/Bob Hunter/Recent/101-0174_IMG.JPG.lnk   Fig.62 Here are all the details of the linked files. .lnk files are used to make shortcuts to a specific location like folder, program. 2, Recycle bin To view all the information of recycle bin to go to this location:

Action	Done?	Date	Time	Notes
				img_Hunter XP for Dongled v6.E01/vol_vol2/RECYCLER/S-1-5-21-1229272821-1580818891-854245398-1004/Df1037  Fig.63 In this folder the suspect has deleted a lot of information shown in the screen shot below. 1, The images about the suspects and victims.

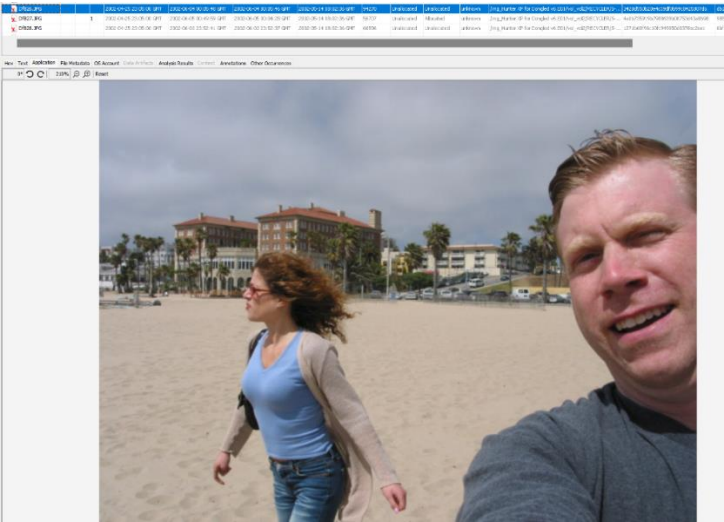
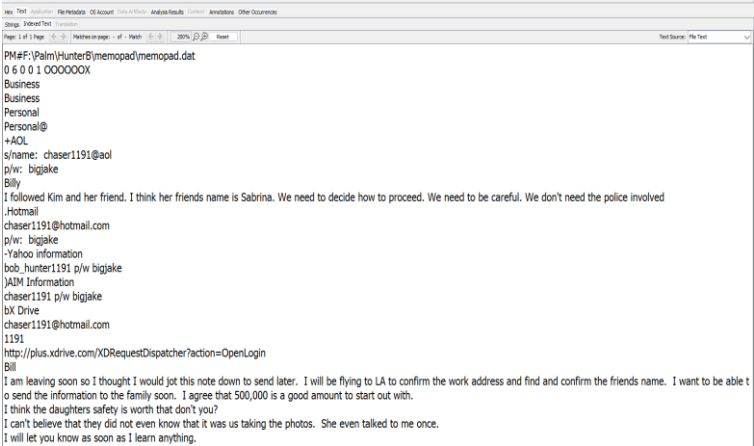
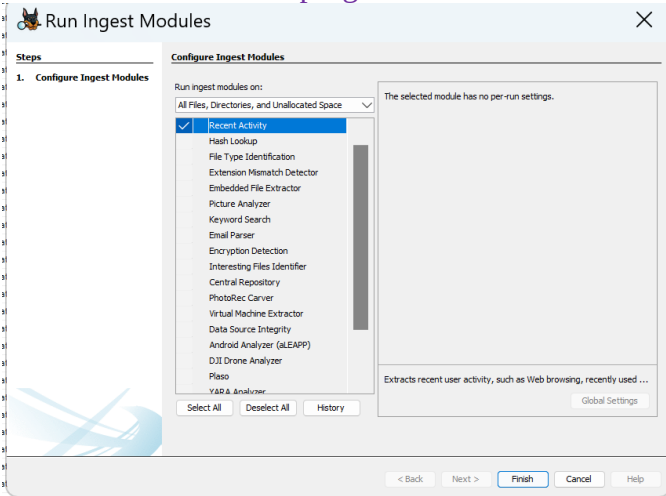
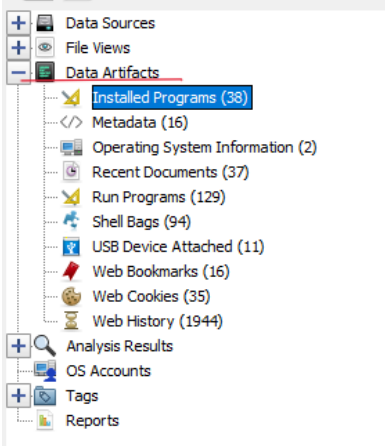
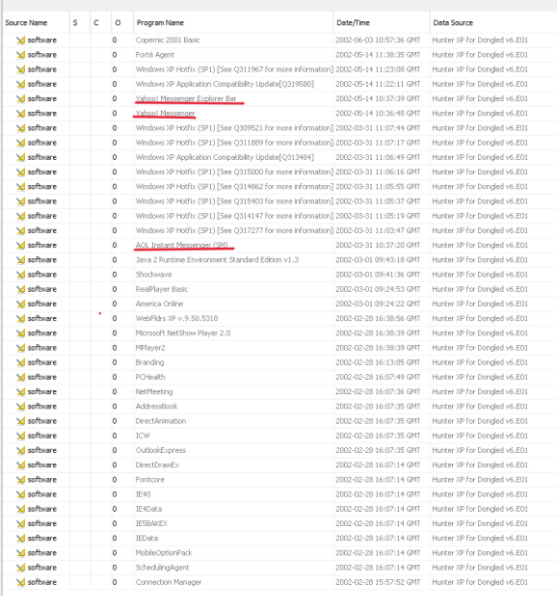
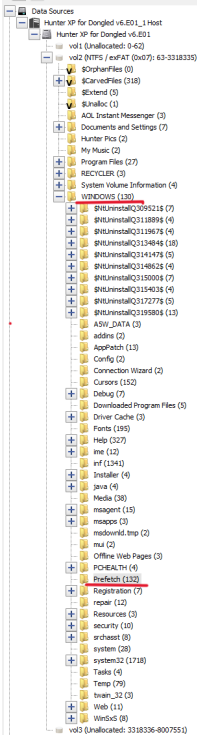
Action	Done?	Date	Time	Notes
				
				2, The information about the suspect's friend and ransomware. 

Fig.64

Fig.65

Action	Done?	Date	Time	Notes
Instant Messaging clients	Yes	27 th November 2024	12:08 Pm	To know the details of instant messaging clients, First, run ingest module “recent activity” Then navigate through: Data artifacts/installed programs/  Fig.66  Fig.67

Action	Done?	Date	Time	Notes
				 Fig.68 Here, these are the 2 messaging clients: 1. Yahoo messenger 2. AOL Instant Messenger (SM)
Clean-up/Wiping utilities. Check log files. Anything used?	Yes	28 th November 2024	2:06 Pm	To see the information regarding the wiping software used navigate through: Prefetch is the location where all the installed application or programs are stored. Hunterxp/windows/prefetch

Action	Done?	Date	Time	Notes
				 Fig.69 After navigating through this location, we can see a software for cleaning data called “cleanmgr.exe”

Action	Done?	Date	Time	Notes																																																																																																						
				<table><thead><tr><th>Name</th><th>S</th><th>C</th><th>O</th><th>Modified Time</th><th>Change Time</th></tr></thead><tbody><tr><td>GLJ3.TMP-06FE9C2B.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:11:52 GMT</td><td>2002-03-01 15:11:52 GMT</td></tr><tr><td>GLJ25.TMP-176642C3.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:26:43 GMT</td><td>2002-03-01 15:26:43 GMT</td></tr><tr><td>GLB98.TMP-26E37750.pf</td><td></td><td></td><td>0</td><td>2002-03-31 16:22:19 GMT</td><td>2002-03-31 16:22:19 GMT</td></tr><tr><td>GLB23.TMP-2405BA7C.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:26:38 GMT</td><td>2002-03-01 15:26:38 GMT</td></tr><tr><td>GLB1B.TMP-1354A202.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:58:36 GMT</td><td>2002-03-01 15:58:36 GMT</td></tr><tr><td>GLB13.TMP-0BAC31D5.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:57:42 GMT</td><td>2002-03-01 15:57:42 GMT</td></tr><tr><td>GLB1.TMP-148E81C1.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:38:59 GMT</td><td>2002-03-01 15:38:59 GMT</td></tr><tr><td>EXTRAS.EXE-3AEFA9A5.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:44:58 GMT</td><td>2002-03-01 15:44:58 GMT</td></tr><tr><td>EXPLORER.EXE-082F38A9.pf</td><td></td><td></td><td>0</td><td>2002-05-14 18:05:13 GMT</td><td>2002-05-14 18:05:13 GMT</td></tr><tr><td>DISKINST.EXE-0E0B1E8F.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:58:21 GMT</td><td>2002-03-01 15:58:21 GMT</td></tr><tr><td>DFRGNTFS.EXE-269967DF.pf</td><td></td><td></td><td>0</td><td>2002-06-03 23:47:22 GMT</td><td>2002-06-03 23:47:22 GMT</td></tr><tr><td>DEFRAG.EXE-273F131E.pf</td><td></td><td></td><td>0</td><td>2002-06-03 23:47:21 GMT</td><td>2002-06-03 23:47:21 GMT</td></tr><tr><td>DAHOTFIX.EXE-0220060F.pf</td><td></td><td></td><td>0</td><td>2002-03-31 16:47:49 GMT</td><td>2002-03-31 16:47:49 GMT</td></tr><tr><td>CONTROL.EXE-013DBFB5.pf</td><td></td><td></td><td>0</td><td>2002-03-31 16:51:10 GMT</td><td>2002-03-31 16:51:10 GMT</td></tr><tr><td>CLEANMGR.EXE-1F86EAE8.pf</td><td></td><td></td><td>0</td><td>2002-06-03 17:08:44 GMT</td><td>2002-06-03 17:08:44 GMT</td></tr><tr><td>CDBOOT.EXE-02410F68.pf</td><td></td><td></td><td>0</td><td>2002-03-01 15:58:21 GMT</td><td>2002-03-01 15:58:21 GMT</td></tr></tbody></table>	Name	S	C	O	Modified Time	Change Time	GLJ3.TMP-06FE9C2B.pf			0	2002-03-01 15:11:52 GMT	2002-03-01 15:11:52 GMT	GLJ25.TMP-176642C3.pf			0	2002-03-01 15:26:43 GMT	2002-03-01 15:26:43 GMT	GLB98.TMP-26E37750.pf			0	2002-03-31 16:22:19 GMT	2002-03-31 16:22:19 GMT	GLB23.TMP-2405BA7C.pf			0	2002-03-01 15:26:38 GMT	2002-03-01 15:26:38 GMT	GLB1B.TMP-1354A202.pf			0	2002-03-01 15:58:36 GMT	2002-03-01 15:58:36 GMT	GLB13.TMP-0BAC31D5.pf			0	2002-03-01 15:57:42 GMT	2002-03-01 15:57:42 GMT	GLB1.TMP-148E81C1.pf			0	2002-03-01 15:38:59 GMT	2002-03-01 15:38:59 GMT	EXTRAS.EXE-3AEFA9A5.pf			0	2002-03-01 15:44:58 GMT	2002-03-01 15:44:58 GMT	EXPLORER.EXE-082F38A9.pf			0	2002-05-14 18:05:13 GMT	2002-05-14 18:05:13 GMT	DISKINST.EXE-0E0B1E8F.pf			0	2002-03-01 15:58:21 GMT	2002-03-01 15:58:21 GMT	DFRGNTFS.EXE-269967DF.pf			0	2002-06-03 23:47:22 GMT	2002-06-03 23:47:22 GMT	DEFRAG.EXE-273F131E.pf			0	2002-06-03 23:47:21 GMT	2002-06-03 23:47:21 GMT	DAHOTFIX.EXE-0220060F.pf			0	2002-03-31 16:47:49 GMT	2002-03-31 16:47:49 GMT	CONTROL.EXE-013DBFB5.pf			0	2002-03-31 16:51:10 GMT	2002-03-31 16:51:10 GMT	CLEANMGR.EXE-1F86EAE8.pf			0	2002-06-03 17:08:44 GMT	2002-06-03 17:08:44 GMT	CDBOOT.EXE-02410F68.pf			0	2002-03-01 15:58:21 GMT	2002-03-01 15:58:21 GMT
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CDBOOT.EXE-02410F68.pf			0	2002-03-01 15:58:21 GMT	2002-03-01 15:58:21 GMT																																																																																																					

Fig.70

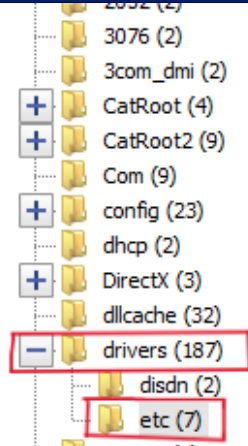
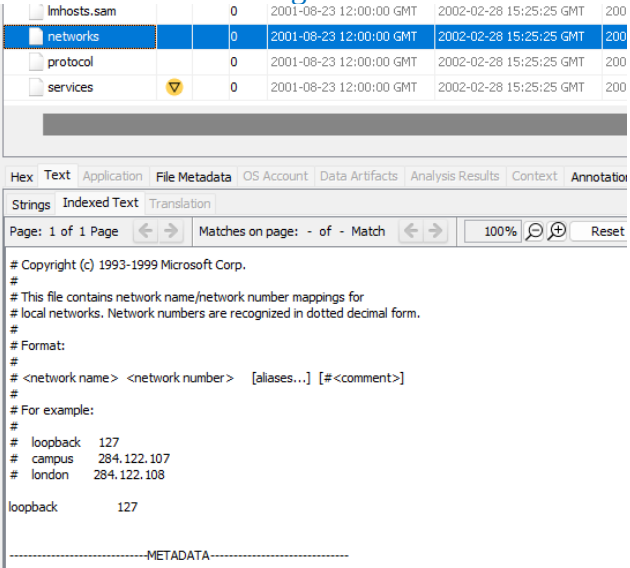
There are also details researched about Clean-up/wiping utilities for this navigate through location:

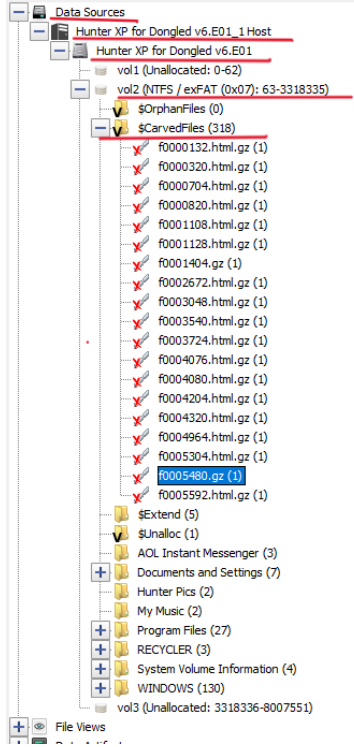
1. Here he researched about BCWIPE tool:

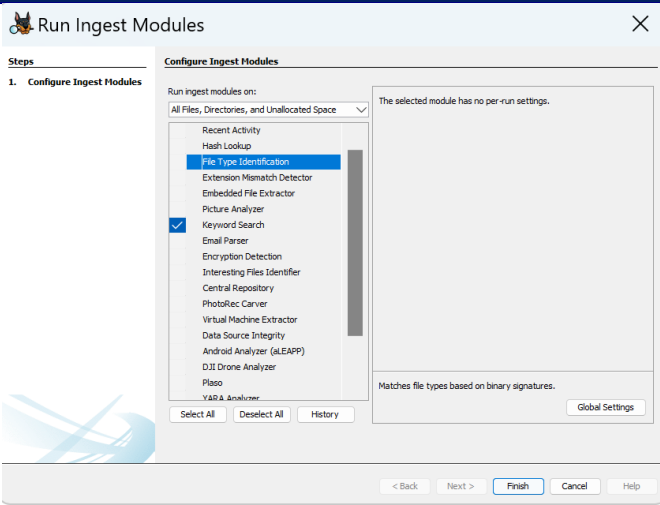
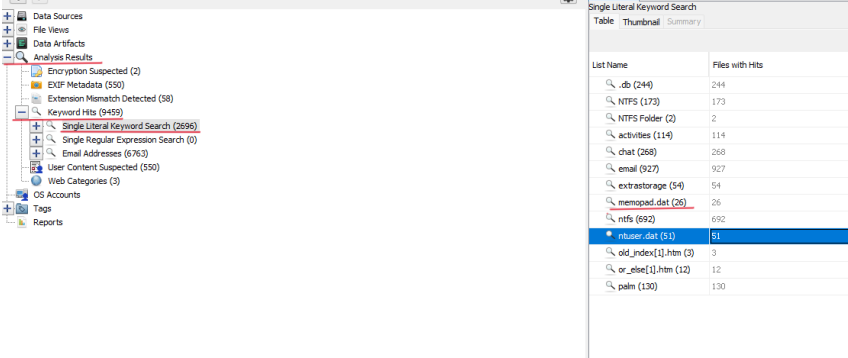
/Img_Hunter XP for Dongled v6.E01/vol_vol2/Document and Settings/Bob Hunter/Local Settings/Temporary Internet Files/Content.IE5/8XGPQD6L/search [3]/search [3]/

Action	Done?	Date	Time	Notes
				1 free space on the specified disk as well as file slacks. Click here to read more and download BCWipe utility 8. The main commands to control access to encrypted data may be run from Windows Explorer ("My Computer" window) without starting BestCrypt Control Panel. To do it you should use the sar 9. BestCrypt has been designed as the product with an open architecture: any third-party encryption software and hardware developers are able to insert their own security extensions into the BestC algorithms. To get additional information about the architecture, see the BestCrypt Development Kit for Windows 95/98/ME/NT/2000/XP WWW-page. 10. The BestCrypt Development Kit (available for downloading from Jetico, Inc. Web site) contains source codes of all BestCrypt modules responsible for security of the software. 11. BestCrypt automatically saves information about sharing its drives in network. Now user does not need to share the BestCrypt drives every time he/she reboots operating system. What's new and what have been fixed over the previous version 1. New 7.00.b beta version of BestCrypt is announced. (07-September-2001 release, BestCrypt Control Panel v.7.00.b, versions of drivers: v.3.12 for Windows 2000/XP, v.2.41 for Windows 95, v. 2. Problem with IRQL_NOT_LESS_OR_EQUAL or FILE_SYSTEM error occurred when Windows 2000/XP reboots is solved. As well problem of mounting network containers is solved. (02-Oct Windows NT) 3. Few enhancements were made in low-level BestCrypt driver for Windows XP (28-October-2001 release, BestCrypt Control Panel v.7.03.b, versions of drivers: v.3.15 for Windows 2000/XP, v.2 4. Serious problem of reading NTFS formatted containers is fixed in low-level BestCrypt driver for Windows XP (12-November-2001 release, BestCrypt Control Panel v.7.03.b, versions of drivers 5. Swap File Encryption utility now supports all encryption algorithms available in BestCrypt (Rijndael, Blowfish, Twofish, GOST). Option to remove additional passwords is implemented. (14-Nov for Windows NT) 6. Commercial release. (20-November-2001 release, BestCrypt Control Panel v.7.05, versions of drivers: v.3.18 for Windows 2000/XP, v.2.41 for Windows 95, v.2.18 for Windows NT) 7. In Windows 2000/NT/XP BestCrypt now supports creating and formatting containers by users with ordinary privileges (i.e. not by Administrators only). As well, during "Log off" process BestC of drivers: v.3.21 for Windows 2000/XP, v.2.41 for Windows 95/98/ME, v.2.18 for Windows NT) 8. In Windows 2000/XP BestCrypt now dismounts virtual drives during restart/shutdown process to perform additional work on flushing cache to NTFS formatted virtual drives. (17-January-2002 Windows NT) 9. In Windows 2000/XP version of BestCrypt problem of mounting several FAT32 formatted containers is solved. In the Windows 95/98/ME version we could catch and solve occasionally arised ; Windows 2000/XP, v.2.41 for Windows 95/98/ME, v.2.18 for Windows NT) 10. In Windows 2000/XP version of BestCrypt problem of saving correct security identifiers on NTFS-formatted containers is solved. (13-March-2002 release, BestCrypt Control Panel v.7.07, v 11. Windows 2000/XP now properly reports user if he/she mounts FAT/FAT32 formatted containers in read-only mode. As well we have inserted new recommendation to the "Hidden containers: ; filesystem if you are going to create hidden part inside it (see more detail in the BestCrypt help file). (06-April-2002 release, BestCrypt Control Panel v.7.07.2, versions of drivers: v.3.27 for Wind If you want to evaluate BestCrypt v.7 for Windows 95/98/ME/NT/2000/XP, visit the Download Centre WWW-page to download the software. Upgrade policy to version 7. All registered customers of any of previous versions 4.x, 5.x or 6.x of BestCrypt software are eligible for a special price to upgrade to version 7. To make an order for upgrade, visit Order page. If you have any questions on upgrade procedure, feel free to contact our Sales Department at sales@jetico.com Fig.73
External drives; Network connections	Yes	28 th November 2024	11:47 Am	To know the details of External drives, navigate through: Tools/runingestmodule/HunterXP/filetype identification Data artifacts/USB Device attached/

Action	Done?	Date	Time	Notes
				Fig.74 Fig.75 Here are many details related to external devices attached or USB device attached. 2, To know about the network connections, go to this location: Hunterxp/vol2/windows/system32/drivers/etc/network

Action	Done?	Date	Time	Notes
				 Fig.77  Fig.78 In this location we can see all the Ip address connected to the pc.

Action	Done?	Date	Time	Notes
Perform data carving				To see the files whose data are carved navigate through: Data source/vol2/carved files  Fig.79 These are the files that are recovered from allocated space or fragment of data.
Run relevant keyword searches. Did you index the evidence file?	Yes	28 th November 2024	12:45 Pm	I have searched many activities using keyword to see the exact location where the keywords are: First run the ingest module and select keyword Search.

Action	Done?	Date	Time	Notes
				 Fig.80 Then navigate through, Analysis results/keyword hits/single keyword searches/  Fig.81 Through the searches the main keyword is “memopad.dat” in this evidence file there are many details of two suspect talking about the victims.

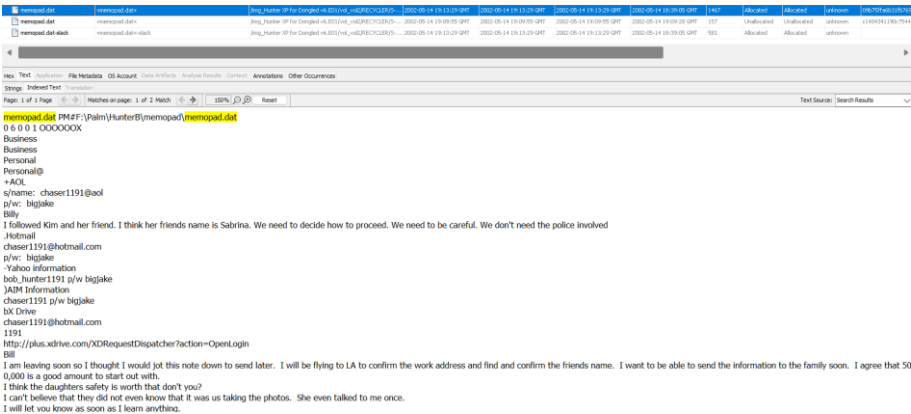
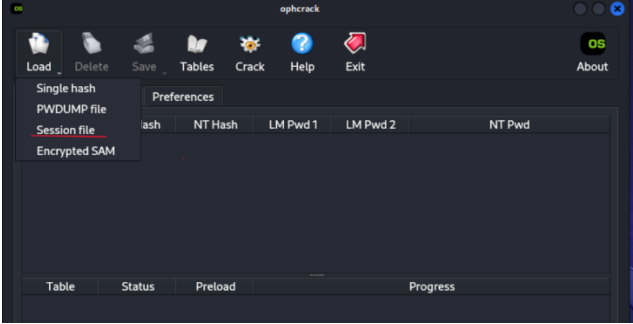
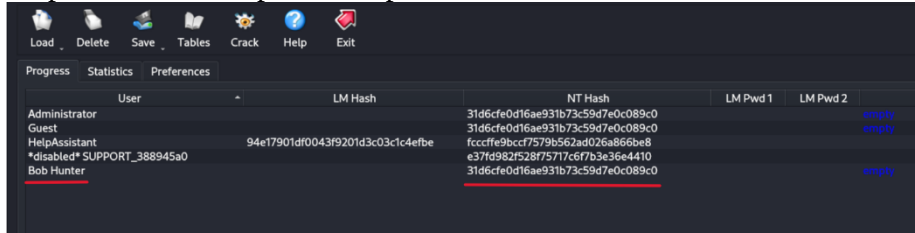
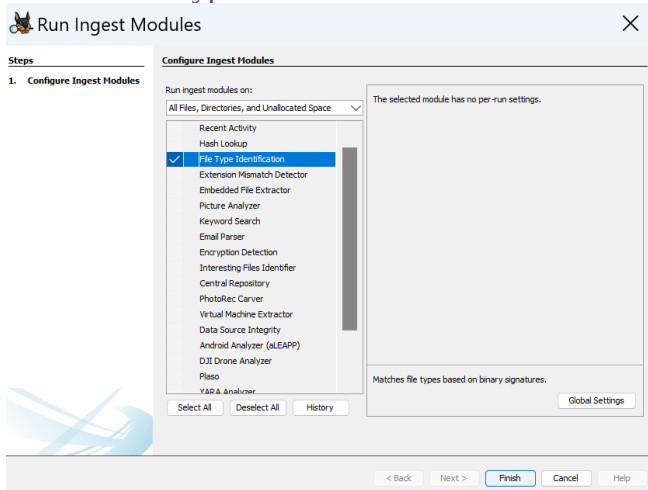
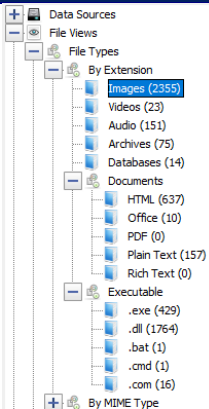
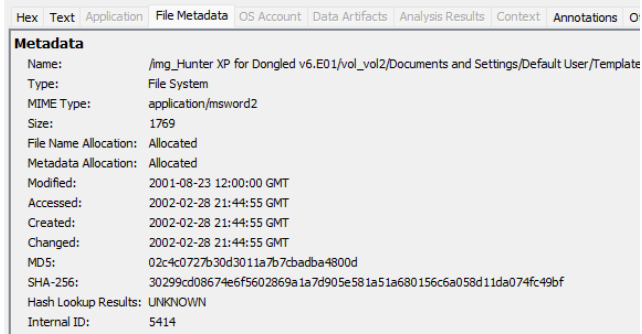
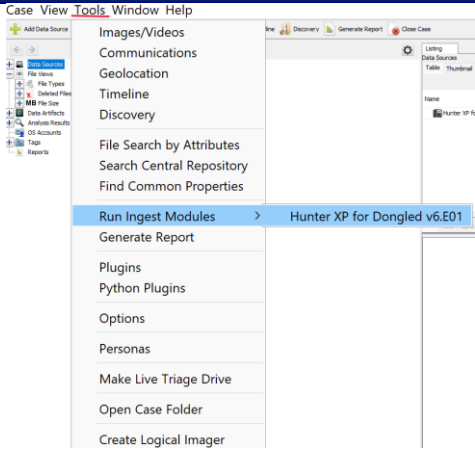
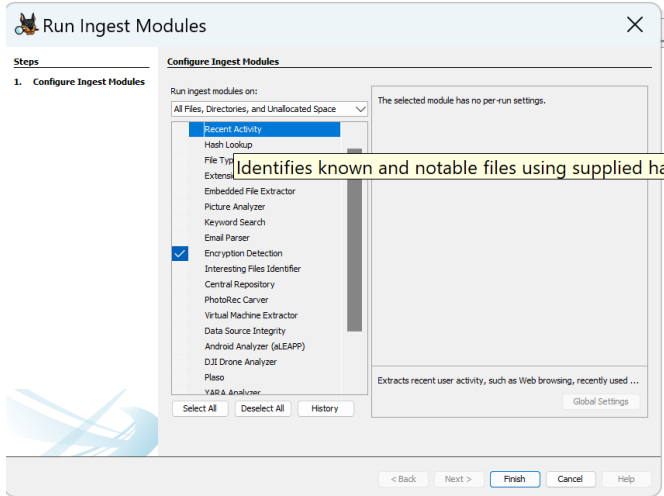
Action	Done?	Date	Time	Notes
				
Recover Log-on passwords – use SAMInside/Ophcrack/Encase				To know about the password of the user first export the config file from: Hunterxp/windows/system32/config Then run the Linux in Virtual machine and open the software Ophcrack. After that just load the config file using the option session file as shown below: 

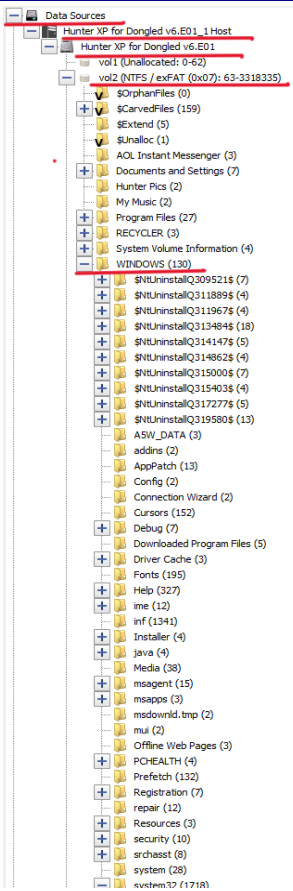
Fig.83

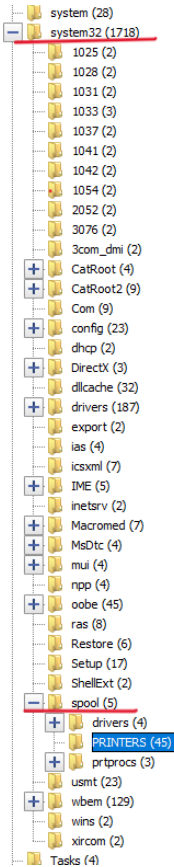
Action	Done?	Date	Time	Notes
				After loading the file, it shows some hash values which gives the output that it is not password protected.  Fig.84
Examine different file types: Export doc/office and exe files; look at Metadata if required	Yes	29 th November 2024	3:04 Pm	To see and examine the details of file types first run the ingest module and then select file type identification after then all the information are generated as shown in the image below: Fileviews/filetypes  Fig.85 Here are different types of files:

Action	Done?	Date	Time	Notes																				
				 Fig.86 Details of Doc files: . Doc are the files containing word documents. <table><tr><td>winword2.doc</td><td>1</td><td>2001-08-23 12:00:00 GMT</td><td>2002-02-28 21:44:55 GMT</td><td>2002-02-2</td></tr><tr><td>winword2.doc</td><td>1</td><td>2001-08-23 12:00:00 GMT</td><td>2002-02-28 22:02:39 GMT</td><td>2002-02-2</td></tr><tr><td>winword.doc</td><td>1</td><td>2001-08-23 12:00:00 GMT</td><td>2002-02-28 21:44:55 GMT</td><td>2002-02-2</td></tr><tr><td>winword.doc</td><td>1</td><td>2001-08-23 12:00:00 GMT</td><td>2002-02-28 22:02:39 GMT</td><td>2002-02-2</td></tr></table>  Fig.87 Details of Exe files: . Exe are those files which contains program or applications.	winword2.doc	1	2001-08-23 12:00:00 GMT	2002-02-28 21:44:55 GMT	2002-02-2	winword2.doc	1	2001-08-23 12:00:00 GMT	2002-02-28 22:02:39 GMT	2002-02-2	winword.doc	1	2001-08-23 12:00:00 GMT	2002-02-28 21:44:55 GMT	2002-02-2	winword.doc	1	2001-08-23 12:00:00 GMT	2002-02-28 22:02:39 GMT	2002-02-2
winword2.doc	1	2001-08-23 12:00:00 GMT	2002-02-28 21:44:55 GMT	2002-02-2																				
winword2.doc	1	2001-08-23 12:00:00 GMT	2002-02-28 22:02:39 GMT	2002-02-2																				
winword.doc	1	2001-08-23 12:00:00 GMT	2002-02-28 21:44:55 GMT	2002-02-2																				
winword.doc	1	2001-08-23 12:00:00 GMT	2002-02-28 22:02:39 GMT	2002-02-2																				

Action	Done?	Date	Time	Notes
				 Case View Tools Window Help - Images/Videos - Communications - Geolocation - Timeline - Discovery - File Search by Attributes - Search Central Repository - Find Common Properties - Run Ingest Modules > Hunter XP for Dongled v6.E01 - Generate Report - Plugins - Python Plugins - Options - Personas - Make Live Triage Drive - Open Case Folder - Create Logical Imager Fig.89  Run Ingest Modules Steps 1. Configure Ingest Modules Run ingest modules on: All Files, Directories, and Unallocated Space Recent Activity - Hash Lookup - File Type - Extensions - Identifies known and notable files using supplied hashes - Embedded File Extractor - Picture Analyzer - Keyword Search - Email Parser - Encryption Detection - Interesting Files Identifier - Central Repository - PhotoRec Carver - Virtual Machine Extractor - Data Source Integrity - Android Analyzer (aLEAPP) - DJI Drone Analyzer - Piles - YARA Analyzer Select All Deselect All History Global Settings < Back Next > Finish Cancel Help Fig.90 After this the details of encryption with justification (entropy) are seen.

Action	Done?	Date	Time	Notes
				Encryption Suspected
				Table Thumbnail Summary
				Source Name S C O Source Type Score Conclusion Configuration Justification Comment
				 oem bios.bin 0 File Likely Notable Suspected encryption due to high entropy (7.999988). Suspected encryption due to high entropy (7.999988).

Action	Done?	Date	Time	Notes
				 Fig.92

Action	Done?	Date	Time	Notes
				 Fig.93 Here we can see all the details related to printers.

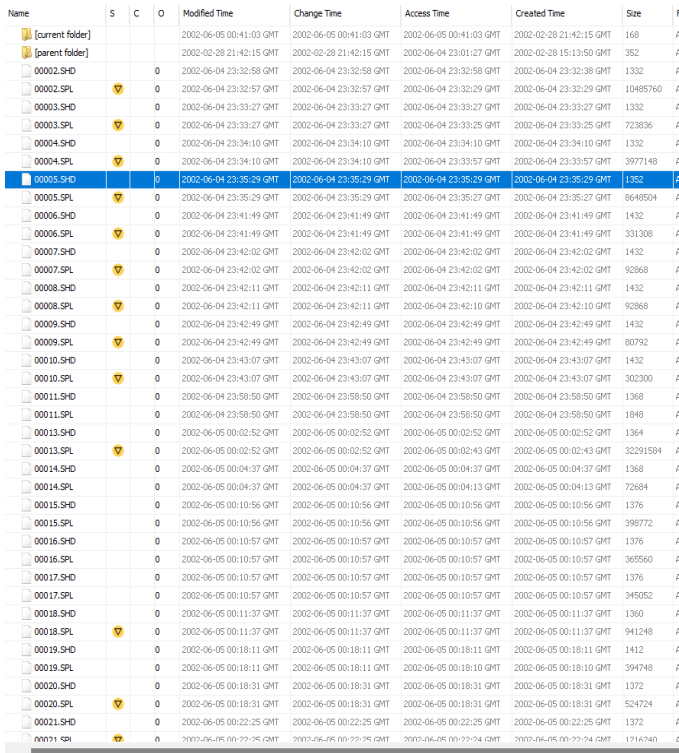
Action	Done?	Date	Time	Notes
				 The screenshot displays a file system directory listing. The columns are: Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, and I. The files listed include folders like '[current folder]' and '[parent folder]', and various files with extensions like .SHD and .SPL. The files are sorted by modified time, showing a sequence of files from 2002-06-05 00:41:03 GMT to 2002-06-05 00:22:25 GMT. The files are organized into a hierarchical structure, with folders like '00001.SHD' and '00001.SPL' containing sub-files.

Fig.94

In this image there are two types of files:

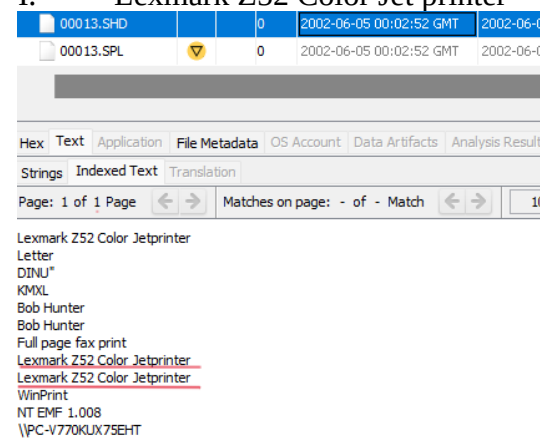
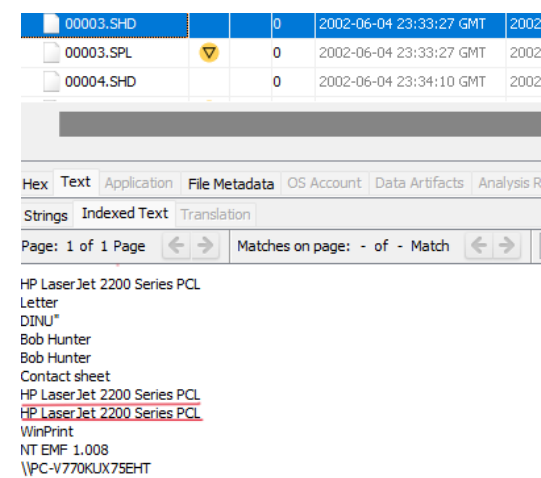
1. . SHD

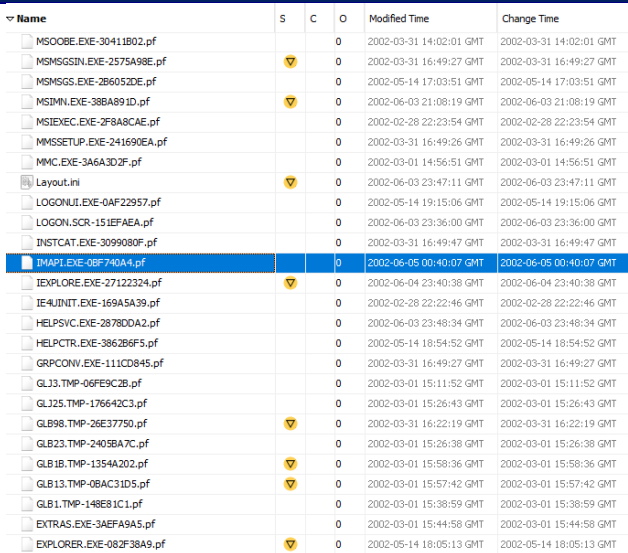
This file contains the details of information the computer sent to printer to execute the process.

2. SPL

This file contains the name of document, owner name, printer name, etc.

Furthermore, there are 2 printers used for this crime:

Action	Done?	Date	Time	Notes
				I. Lexmark Z52 Color Jet printer  Fig.95 ii, HP LaserJet 2200 Series PCL  Fig.96

Action	Done?	Date	Time	Notes
				 Fig.98
Validate evidence integrity at the end of the examination	Yes	30 th November 2024	8:45 Pm	As I finished the investigation, I need to validate the hash value if the data is altered or not. To validate the hash value, I will use AUTOPSY TOOL after that I will match the initial hash value and ending hash value. INITAIL HASH VALUE FROM AUTOPSY 4.19.2:

Additional Notes/Artefacts Examined:

Colour-coding Legend	Tasks
	Fundamental
	Basic
	Elementary
	Secondary
	Advanced
	Exceptional